

CSE 430 Architectural Design

Career Bound

Software Design Document

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Revision History

Version	Date	Author	Change Description
1.0	5/14/26	Jeremy Troy Suchanski	Filled in work done for Milestone 1 and started filling in sections relevant to Milestone 2 for Initial SWDD draft
2.0	5/22/26	Nathan Lipovich	Filled in the diagrams for 6.2 and 7 as well as the 6.1 introduction
3.0	5/24	Corbin Probasco	Fill in the missing information for Component Diagram Design Information Tables & other fixes.

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1.0 INTRODUCTION

1.1 Purpose

This Software Design Document (SWDD) describes the architecture, subsystem breakdown, data design, component structure, and human interface design of the Career Bound system. The document provides the design foundation necessary for implementation and integration of the Career Bound software platform.

The intended audience of this document includes:

- Developers
- Designers
- QA reviewers
- Project managers
- Instructors
- Stakeholders

This document serves as the primary design reference for the Career Bound project and establishes standardized architecture, interface, workflow, and UI design patterns used throughout the system.

1.2 Scope

Career Bound is a web and mobile application for career networking and job searching designed to connect students and employers through profiles, resumes, job postings, notifications, and search capabilities.

The system supports:

- Student profile management
- Company profile management
- Resume creation and submission
- Job posting management
- Candidate and job search
- Notifications and news feed updates
- Responsive desktop and mobile interfaces

The goals of the project are to:

- Simplify finding jobs and candidates
- Improve communication between students and employers
- Provide a centralized career development platform
- Support scalable and modular software architecture

- Demonstrate standardized software engineering practices

The project delivers the following key benefits:

- Improved organization of career-related information
 - All student profiles, resumes, job postings, company information, news, and notifications are centralized in one easy-to-access platform instead of being scattered across emails, spreadsheets, and multiple websites.
- Streamlined resume and application workflows
 - Students can create, update, and submit resumes quickly and efficiently, while companies can easily review candidates and manage job postings, which reduces the manual work and speeds up the entire hiring process.
- Improved accessibility on desktop and mobile devices
 - Students and companies can access all features (profiles, search, notifications, resumes, etc.) conveniently from any device. The platform provides consistent, high-quality experiences whether on a desktop computer, laptop, tablet, or mobile phone.
- Modular architecture supporting maintainability and scalability
 - The system is built using well-defined and independent components (as shown in the component diagram). This modularity makes the application easier to maintain, update, and expand in the future without disrupting the entire system.

1.3 Overview

This document is organized into the following major sections:

- Introduction and project overview
- System architecture and subsystem decomposition
- Data design and storage structure
- Component design
- Human interface design and screen views
- Requirements traceability
- Supporting appendices and standards

The SWDD combines architectural diagrams, interface views, workflow standards, and UI design patterns to provide a complete description of the Career Bound software system.

1.4 Reference Material

The following references were used in the design of Career Bound:

- Software Requirements Specification (SRS)
- IEEE 1016 Software Design Description guidance
- Course textbook Chapter 4.1 – Design Description
- GitHub CareerBound-SDD Repository Standards and Workflow Documentation

- Career Bound Component Diagram Standards
- Career Bound UI and Wireframe Standards

1.5 Definitions and Acronyms

Term / Acronym	Definition
API	Application Programming Interface
UI	User Interface
UX	User Experience
QA	Quality Assurance
IC	Individual Contributor
SWDD	Software Design Document
SRS	Software Requirements Specification
REST	Representational State Transfer
Wireframe	Simplified UI layout representation
Component Diagram	Diagram representing major system components and interfaces
Responsive Design	UI design adapting to multiple screen sizes

2.0 SYSTEM OVERVIEW

Career Bound is designed as a modular career networking platform that supports interaction between students and employers. The system provides profile management, resume handling, job posting functionality, notifications, and advanced search features.

The system is divided into multiple major services and interface layers in order to improve maintainability, scalability, and separation of responsibilities.

The primary system users include:

- Students
- Employers / Companies
- Administrators

The system supports both desktop and mobile user interfaces through responsive UI design patterns.

Major functional areas include:

- Authentication and user management
- Profile management
- Resume services
- Job posting services
- Search services
- Communication services
- Data storage services

3.0 SYSTEM ARCHITECTURE

3.1 Architectural Design

The Career Bound system is designed using modular, component-based architecture to clearly separate concerns and thus enable scalability and maintainability. The overall system is decomposed into well-defined, replaceable components that communicate through explicit interfaces and ports.

Figure 1.1 below shows the high-level component diagram for the Career Bound system. This diagram shows how the system responsibilities have been partitioned into major subsystems and how those subsystems collaborate to deliver the complete functionality required by the SRS.

Figure 1.1 – Career Bound System Master Component Diagram (UML 2.0)

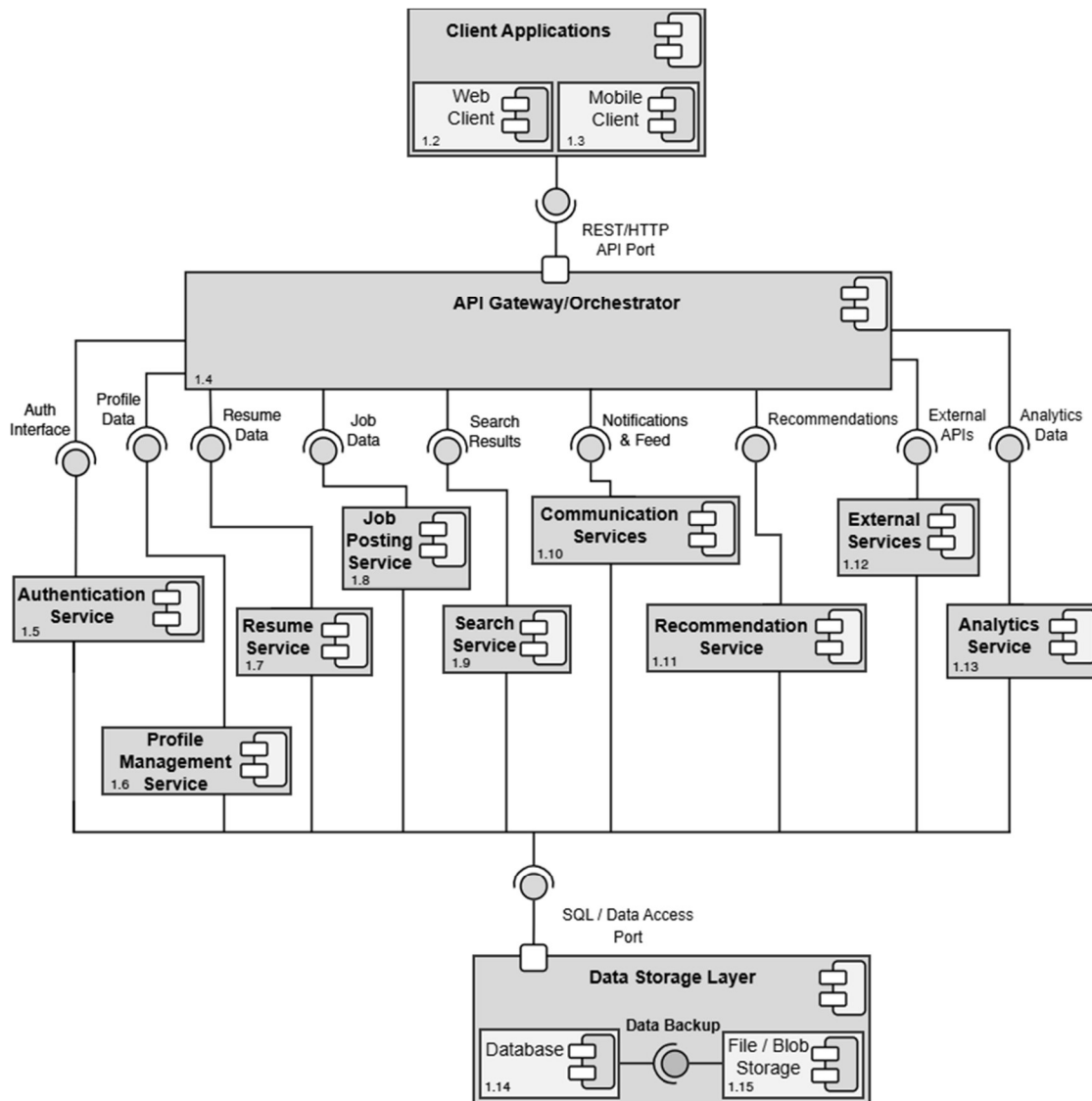


FIGURE 1.1

Design Information

Field	Description / Purpose
View Name & Number	1.1 – Career Bound System Master Component Diagram (UML 2.0)
Purpose	Describe the overall system view.
Description	Basic client, application, database architecture with two types of clients: mobile and web.
Requirements	All SRS sections — 2.1 Company Profile, 2.2 Notifications, 2.3 Search, 2.4 News Feed, 2.5 Job Postings, 3.1 Student Profile, 3.2 Notifications, 3.3 Search, 3.4 News Feed, 3.5 Resume, 4.0 Hardware.
Elements	Client Applications: Web Client and Mobile Client
	API Gateway/Orchestrator
	Auth Service
	Profile Management Service
	Resume Service
	Job Posting Service
	Search Service
	Communications Service
	Recommendation Service
	External Services
	Analytics Service
Navigation / References	Figures 1.2–1.15 all component detail views
Viewpoint	Component Diagram

The interfaces and ports shown in Figure 1.1 are formally defined in the Interface Glossary below. All components communicate exclusively through these explicit interfaces, ensuring low coupling and high replaceability.

Interface Glossary – Career Bound System Master Component Diagram

Interface Name	Description	Provided By	Required By
REST/HTTP API Port	Formal, well-defined entry point for all client (Web + Mobile) communication	API Gateway / Orchestrator	Client Applications (Web Client + Mobile Client)
Auth Interface	Login, registration, role management, and session handling	Authentication Service	API Gateway
Profile Data	Read/write access to company and student profiles (locations, following, skills, GPA, etc.)	Profile Management Service	API Gateway, Search Service, Recommendation Service
Resume Data	Resume creation, submission, storage, retrieval, and access control	Resume Service	API Gateway, Search Service, Company profiles

Interface Name	Description	Provided By	Required By
Job Data	Job posting creation, editing, deletion, required skills, and job details	Job Posting Service	API Gateway, Search Service, Recommendation Service
Search Results	Candidate and job search results, filters, autocomplete, and wildcards	Search Service	API Gateway
Notifications & Feed	In-app notifications, push notifications, news feed posts, and updates	Communication Services	API Gateway
Recommendations	Job and candidate recommendations based on profiles, resumes, and job data	Recommendation Service	API Gateway
External APIs	Integration with third-party services (USPS address validation, autocomplete APIs, push notification providers, etc.)	External Services	API Gateway
Analytics Data	System usage metrics, reporting, and analytics data	Analytics Service	API Gateway
SQL / Data Access Port	Formal port — the single standardized interface through which all services access persistent data (the only way any service talks to the database or file storage)	Data Storage Layer	All services

The architecture is organized into four primary layers:

- **Client Applications (parent)** – Encloses the Web Client and Mobile Client. These components provide the unified user interface for both students and companies, supporting all features described in SRS sections 2.x and 3.x.
- **API Gateway / Orchestrator** – Serves as the single, centralized entry point for all client requests. It routes traffic to the appropriate backend services and enforces cross-cutting concerns such as authentication and request orchestration.
- **Core Services Layer** – Contains the business logic components:
 - Authentication Service
 - Profile Management Service
 - Resume Service
 - Job Posting Service
 - Search Service
 - Communication Services (Notifications + News Feed)
 - Recommendation Service
 - External Services

- Analytics Service

Each service is responsible for a cohesive set of related functions and exposes well-defined interfaces to the rest of the system.

- **Data Storage Layer (parent)** – Acts as the single source of truth for all persistent data. It encloses the Database and File / Blob Storage components and exposes the formal SQL / Data Access Port used by every service in the system.

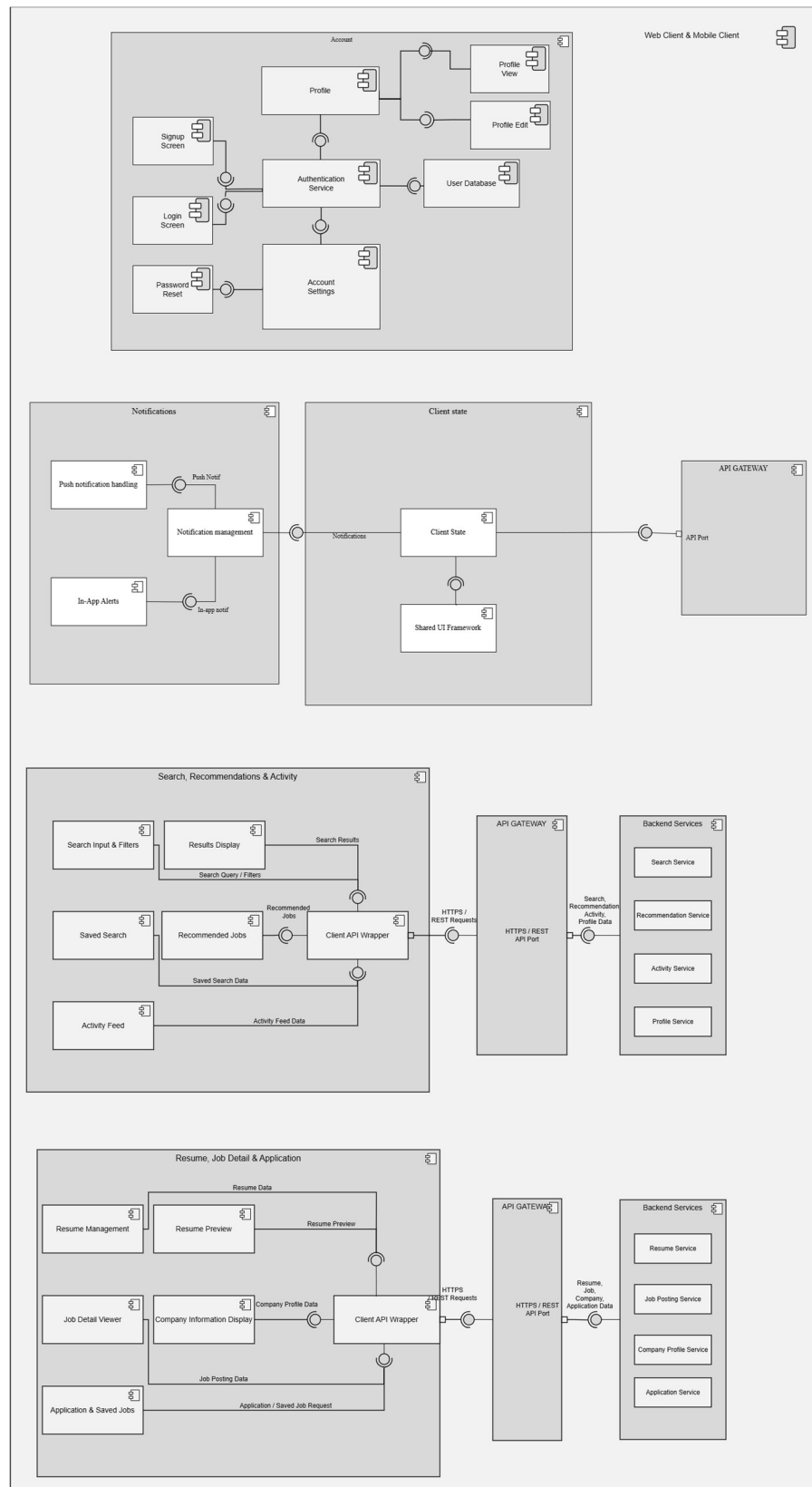
All communication between the Client Applications and the backend services flows through the **REST/HTTP API Port** provided by the API Gateway. Every service communicates with the Data Storage Layer exclusively through the **SQL / Data Access Port**. This design ensures reduced dependencies between components, clear separation of responsibilities, and makes individual components replaceable and maintainable. Detailed views of each major component are provided in Figures 1.2 through 1.15. This modular architecture directly supports the system's requirements for accessibility, scalability, and maintainability while providing a clear roadmap for implementation and future extension.

3.2 Decomposition Description

The master component diagram (Figure 1.1) provides a high-level view of the overall system architecture. This section presents the detailed decomposition of each major component identified in the master diagram. Each sub-diagram shows the internal structure of the component, its sub-components, and the interfaces through which it interacts with the rest of the system. All sub-diagrams maintain the same interface names and ports as defined in the master diagram and the Interface Glossary to ensure consistency.

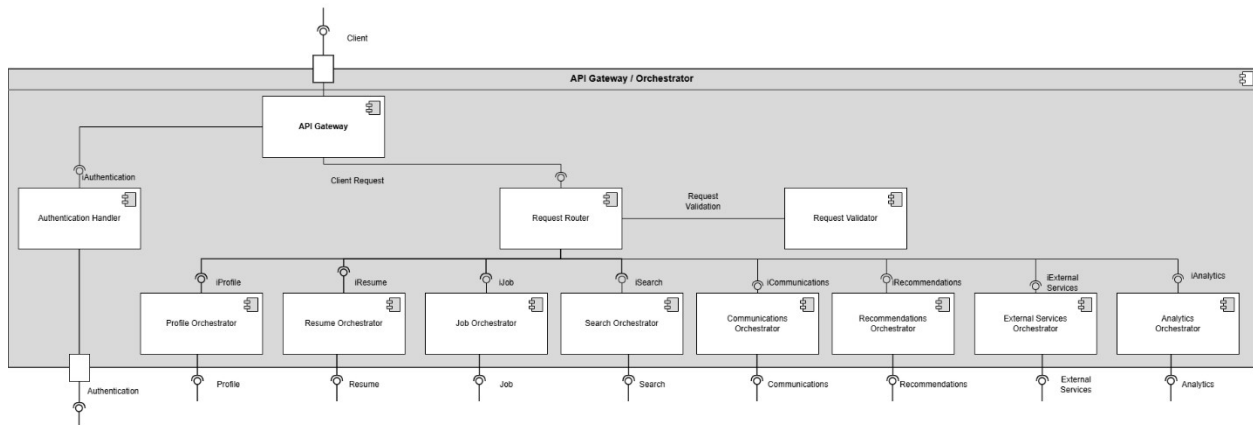
The following figures provide the detailed decomposition of each major component shown in the master diagram (Figure 1.1):

Figure 1.2 & 1.3 – Web & Mobile Client Detail

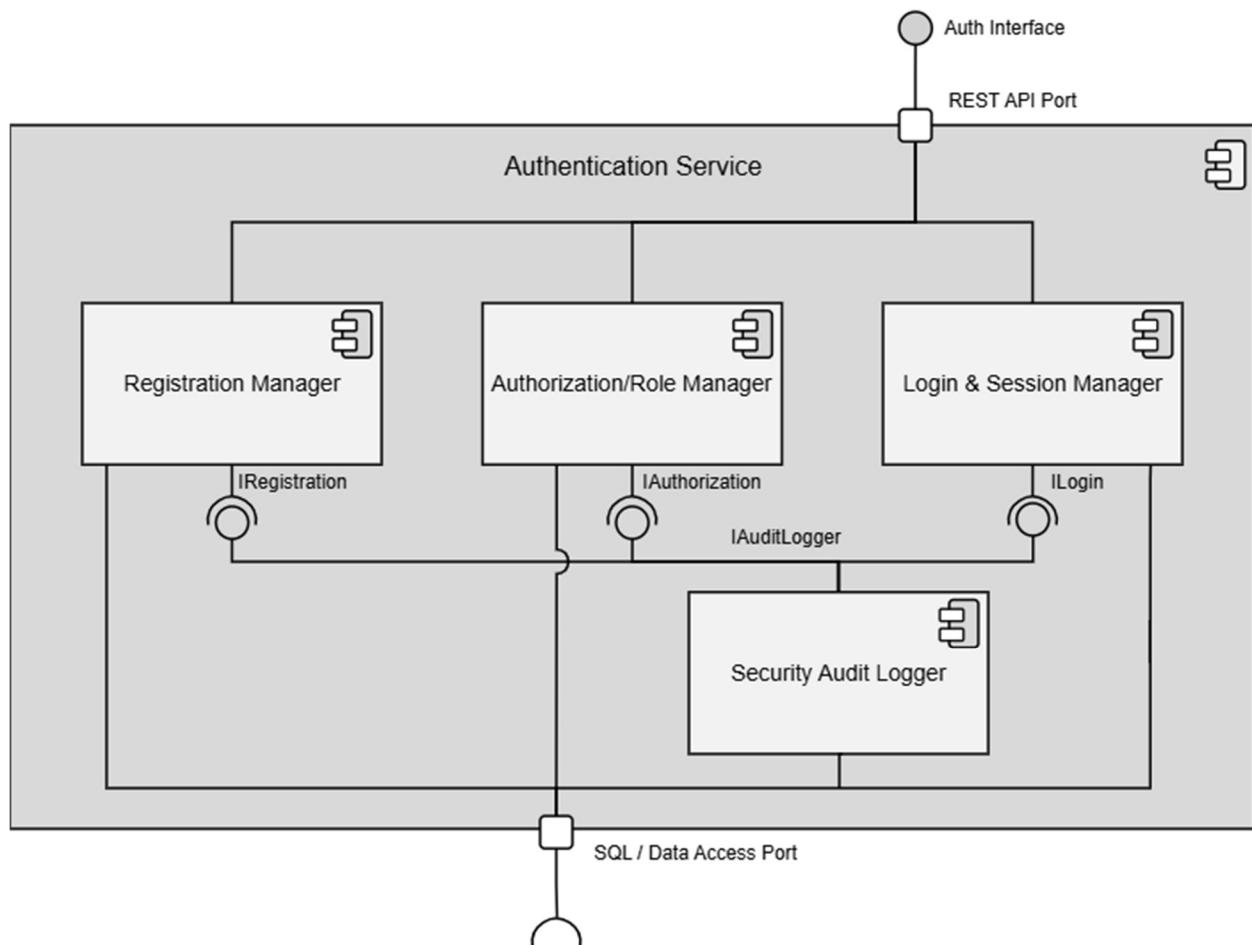


Design Information

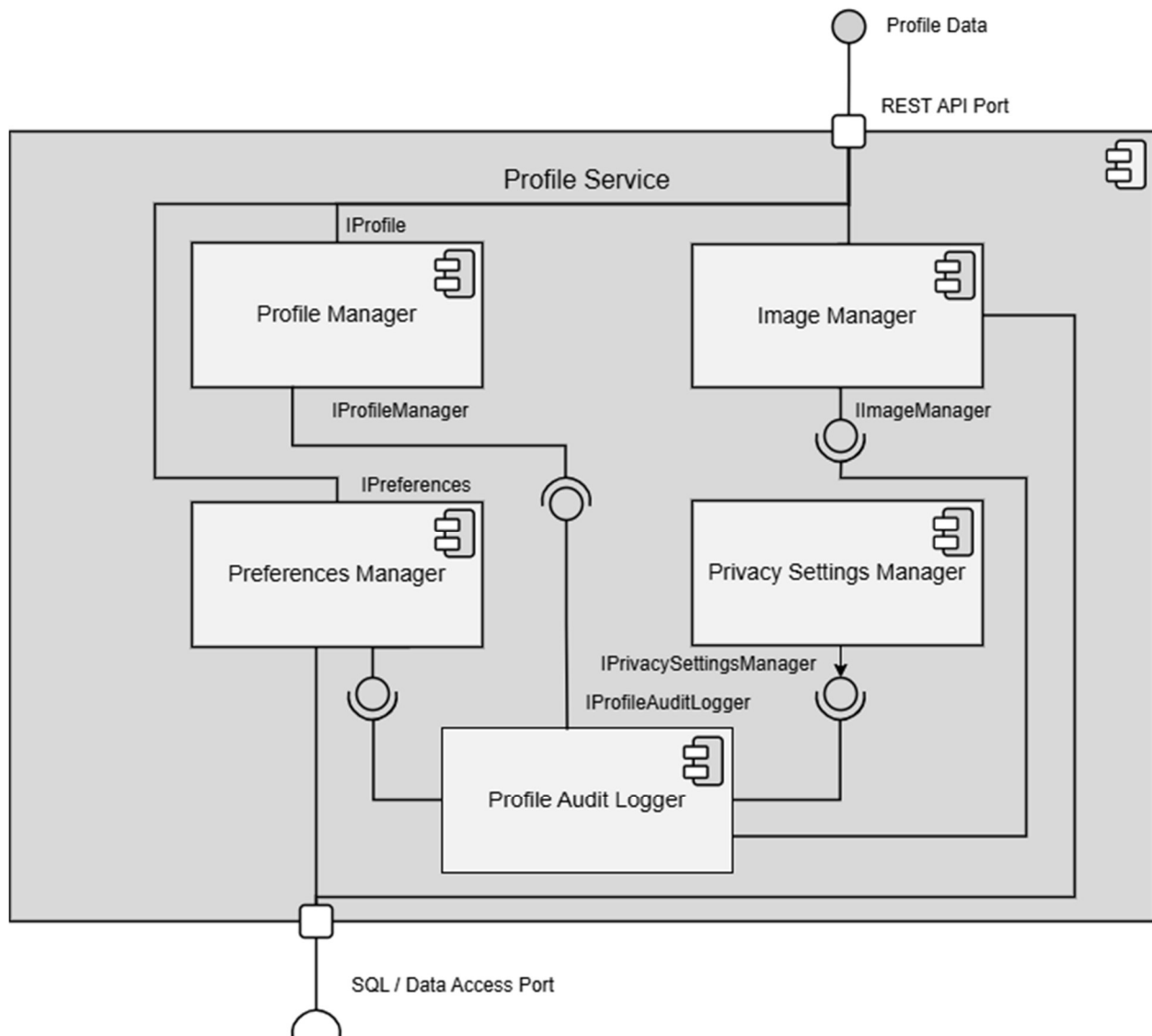
Field	Description / Purpose
View Name & Number	1.2 – Web Client & 1.3 – Mobile Client
Purpose	Describe the overall system view.
Description	Basic client, application, database architecture with two types of clients: mobile and web.
Requirements	Account
Elements	Profile
	Profile View
	Profile Edit
	Authentication Service
	Signup Screen
	Login Screen
	Password Reset
	Account Settings
	User Database
	Notifications
	Notification Management
	Push Notification Handler
	In-App Alerts
	Client State
	Shared UI Framework
	Search, Recommendations, & Activity
	Search Input Filters
	Results Display
	Saved Search
	Recommended Jobs
	Activity Feed
	Client API Wrapper
	Resume Management
	Resume Preview
	Job Detail Viewer
	Company Information Display
	Application & Saved Jobs
	Client API Wrapper
Navigation / References	1.4 API Gateway / Orchestrator; all screen views (1.1–4.2)
Viewpoint	Component Diagram

Figure 1.4 – API Gateway / Orchestrator Detail**Design Information**

Field	Description / Purpose
View Name & Number	1.4 – API Gateway / Orchestrator
Purpose	Describe the API/ Orchestrator View
Description	API Gateway with authentication Handling and request routing to various component orchestration.
Requirements	Cross-cutting for all SRS sections — routes all client requests to backend services
Elements	API Gateway Authentication Handler Auth Service Request Handler Request Validator Profile Orchestrator Resume Orchestrator Job Orchestrator Search Orchestrator Communications Orchestrator Recommendation Orchestrator External Orchestrator Analytics Orchestrator
Navigation / References	1.2 Web Client, 1.3 Mobile Client, 1.5–1.13 all backend service detail views
Viewpoint	Component Diagram

Figure 1.5 – Authentication Service Detail**Design Information**

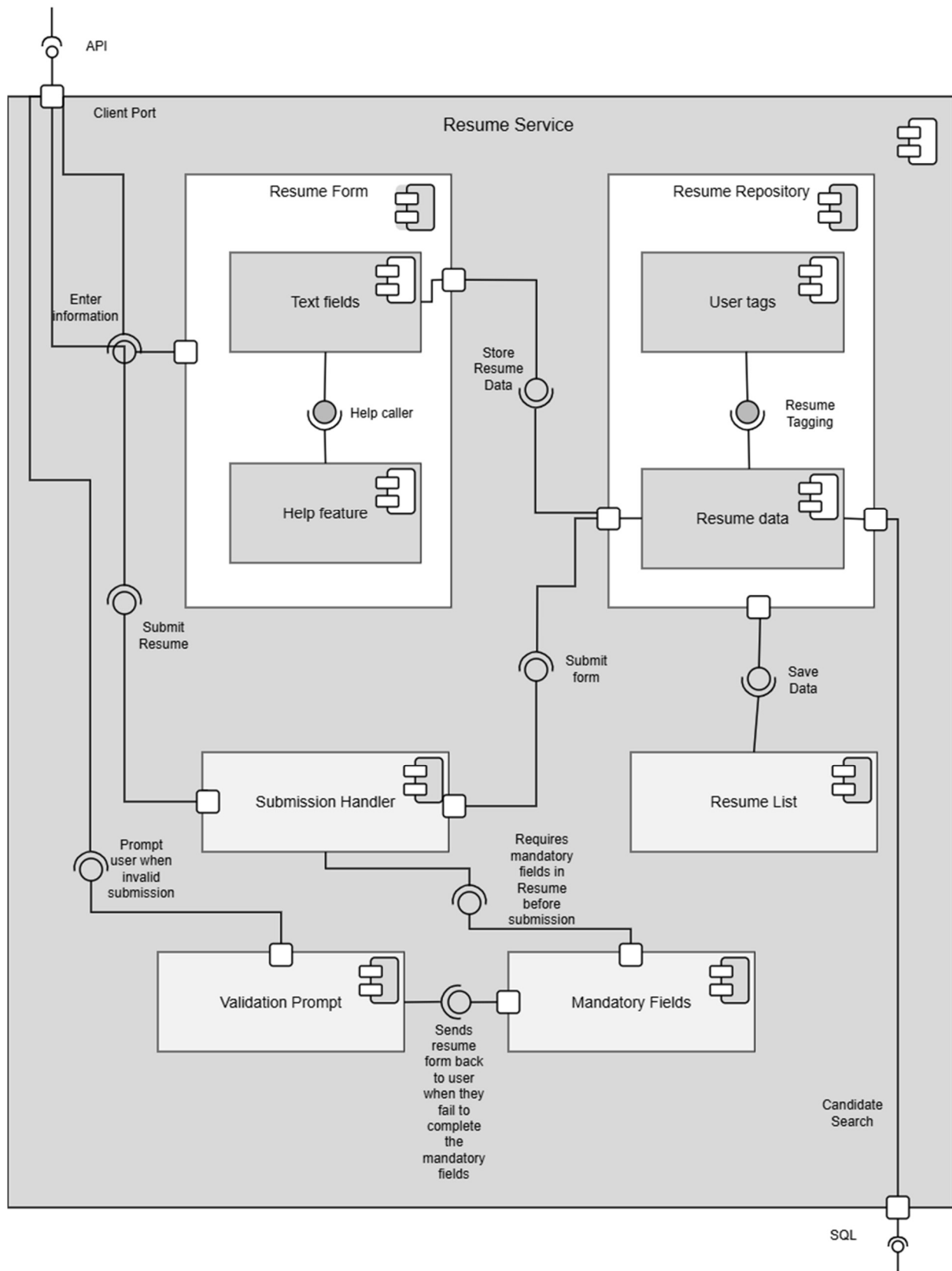
Field	Description / Purpose
View Name & Number	1.5 Authentication Service
Purpose	Describe the authentication system view.
Description	Basic service demonstrating the authentication service.
Requirements	Foundational requirement for all logged-in features — SRS 2.1.3.10 (Security), 3.1.1 (Student account), 2.1.1 (Company account)
Elements	Registration Manager Authorization/ Role Manager Login and Session Manager
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 1.1 Login / Logout, 1.2 Registration
Viewpoint	Component Diagram

Figure 1.6 – Profile Management Service Detail**Design Information**

Field	Description / Purpose
View Name & Number	1.6 Profile Management Service
Purpose	Describe the profile management service.
Description	Basic service demonstrating the profile management service.
Requirements	SRS 2.1 Company Profile Page (2.1.1–2.1.3.10), 3.1 Student User Profile (3.1.1–3.1.3.3)
Elements	Profile Manager Image Manager Preferences Manager Privacy Settings Manager Profile Audit Logger

Field	Description / Purpose
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database, 1.15 File / Blob Storage
Viewpoint	Component Diagram

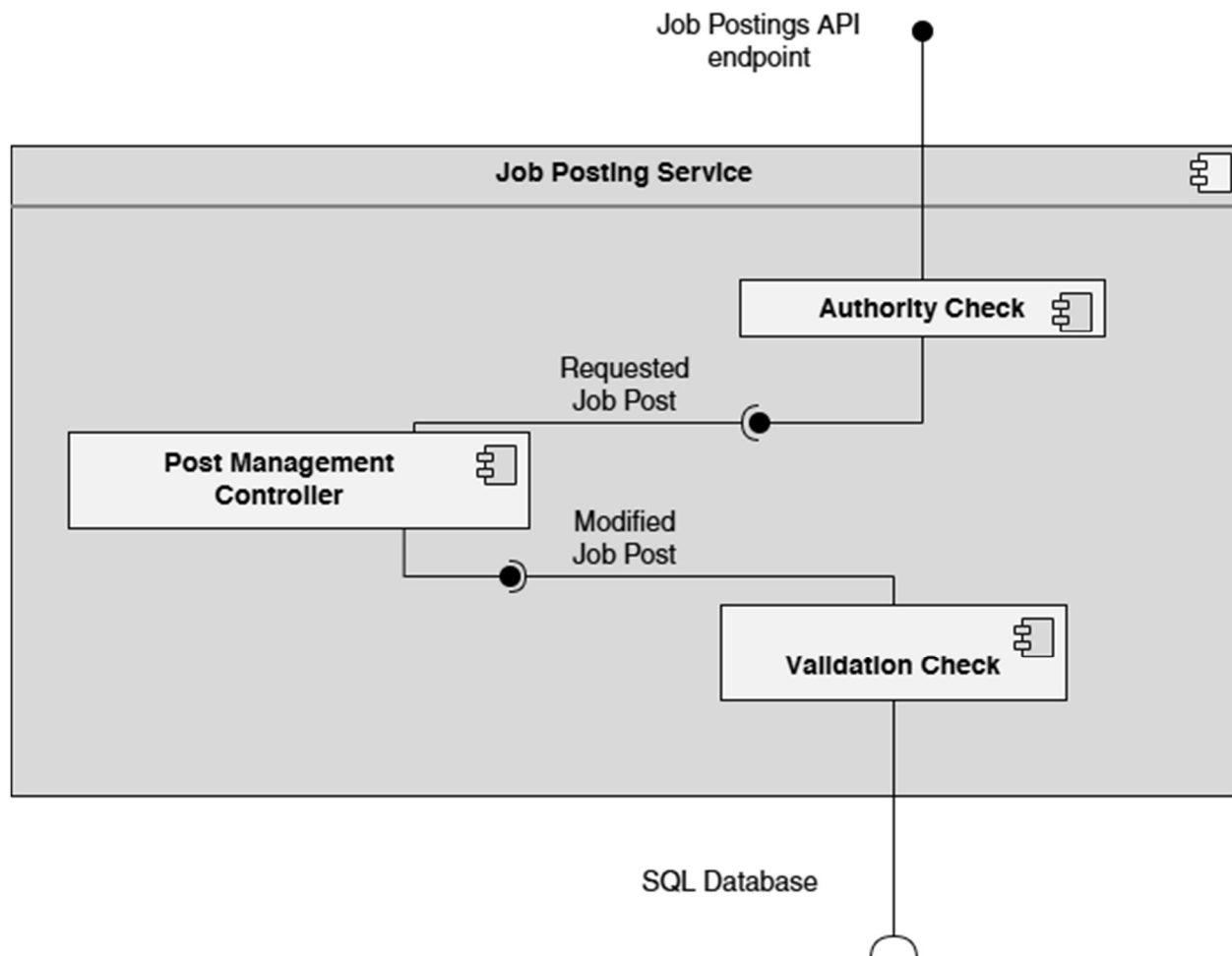
Figure 1.7 – Resume Service Detail



Design Information

Field	Description / Purpose
View Name & Number	1.7 Resume Service
Purpose	Describe the resume service.
Description	Basic service demonstrating the resume service.
Requirements	SRS 3.5 Resume (3.5.1–3.5.5.2); referenced in 2.1.2, 2.2.5, 2.3.3, 2.3.5, 2.5.6
Elements	Resume Form
	Resume Repository
	Text Fields
	Help Feature
	User Tags
	Resume data
	Submission Handler
	Resume List
	Validation Prompt
	Mandatory Fields
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database, 1.15 File / Blob Storage; Screen view 2.3 Resume Builder
Viewpoint	Component Diagram

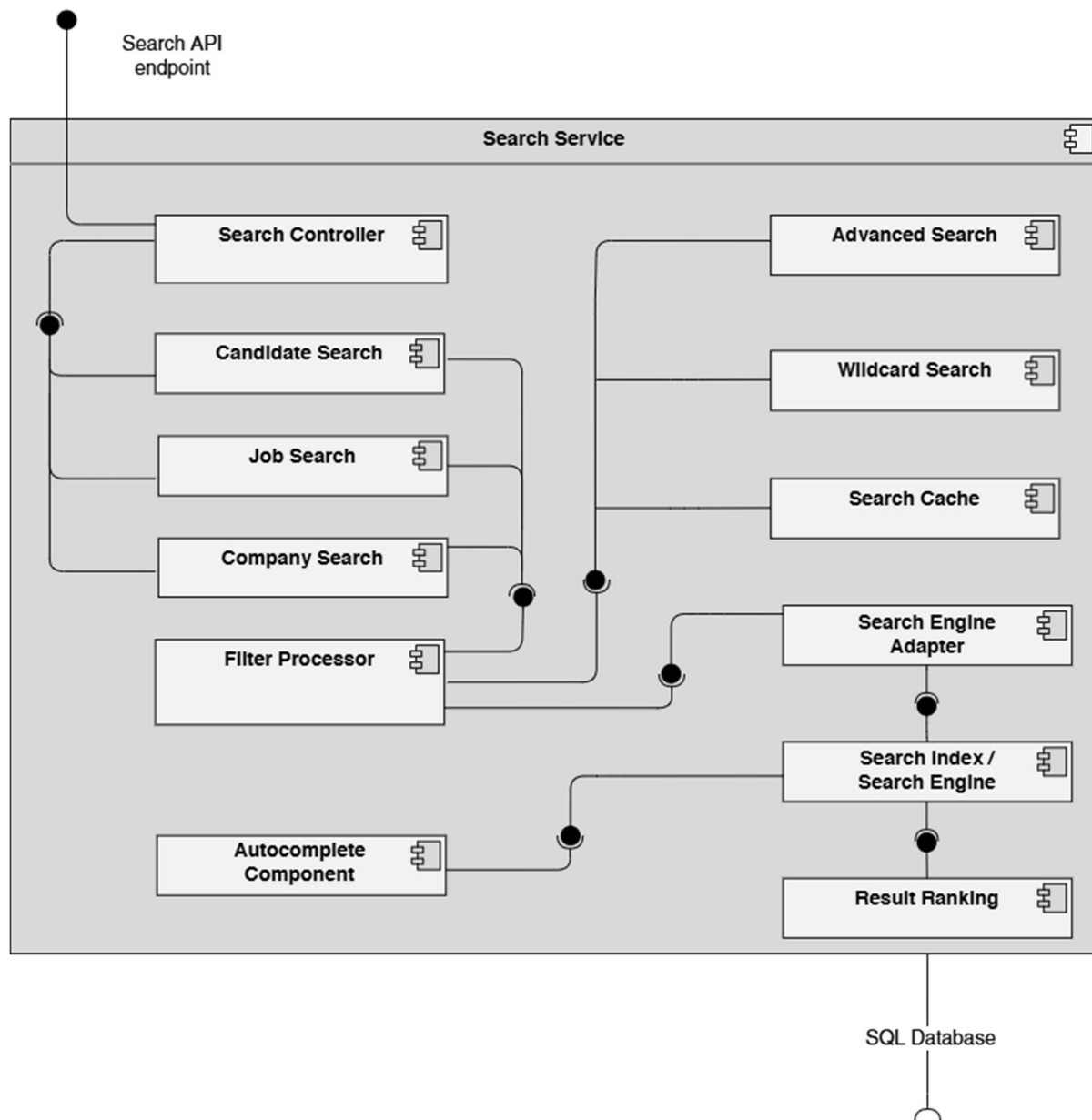
Figure 1.8 – Job Posting Service Detail



Design Information

Field	Description / Purpose
View Name & Number	1.8 Job Postings Service
Purpose	Describe the jobs posting service.
Description	Basic service demonstrating the jobs posting service.
Requirements	SRS 2.5 Job Postings (2.5.1–2.5.7); referenced in 3.3.1, 3.4.1, 2.2.5
Elements	Authority Check Post Management Control Validation Check
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 3.3 Job Posting Creation, 2.4 Job Search, 2.5 Job Detail
Viewpoint	Component Diagram

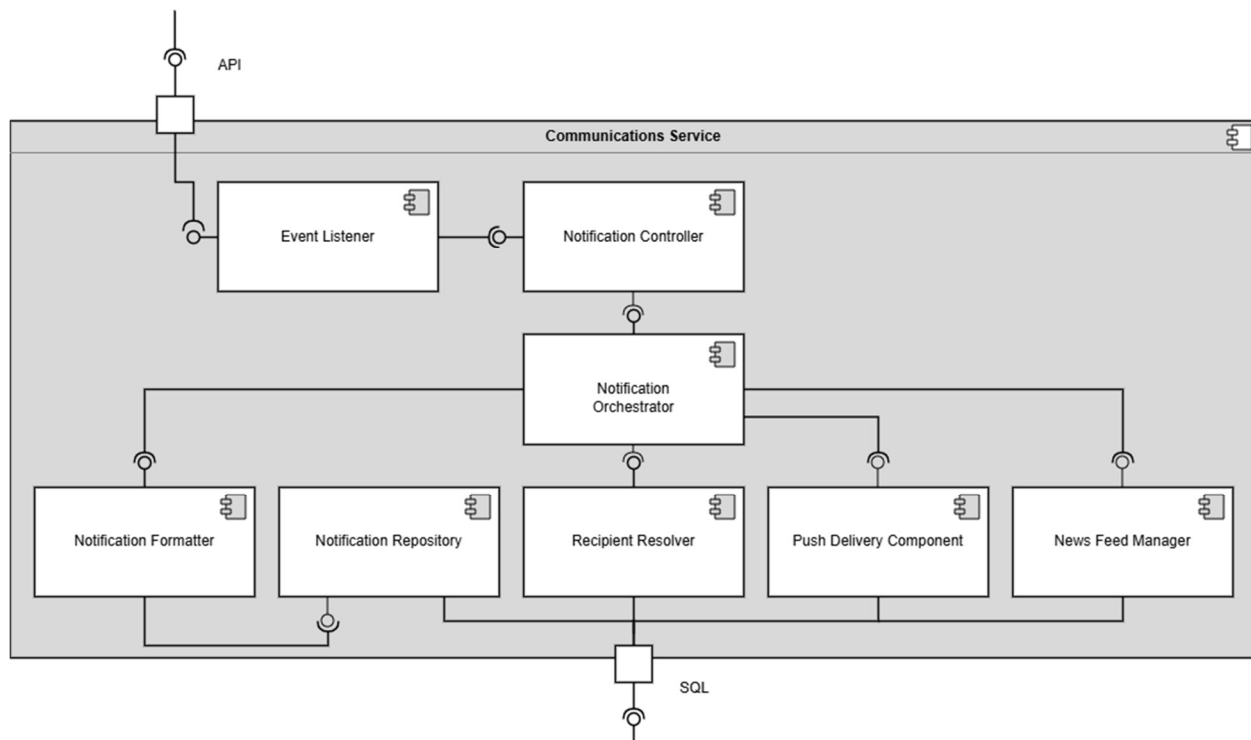
Figure 1.9 – Search Service Detail



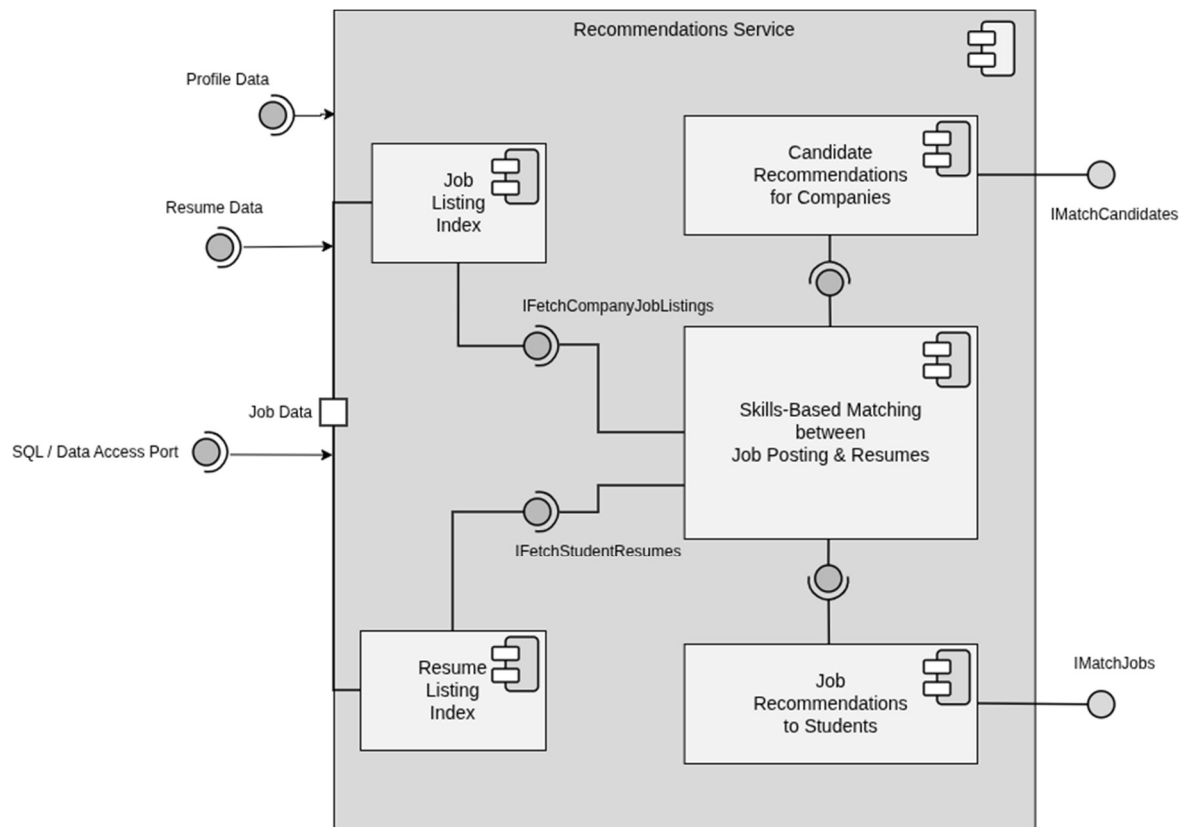
Design Information

Field	Description / Purpose
View Name & Number	1.9 Search Service
Purpose	Describe the search service.
Description	Basic service demonstrating the search service.
Requirements	SRS 2.3 Search Features (2.3.1–2.3.8), 3.3 Student Search Features (3.3.1–3.3.2)
Elements	Search Controller
	Advanced Search

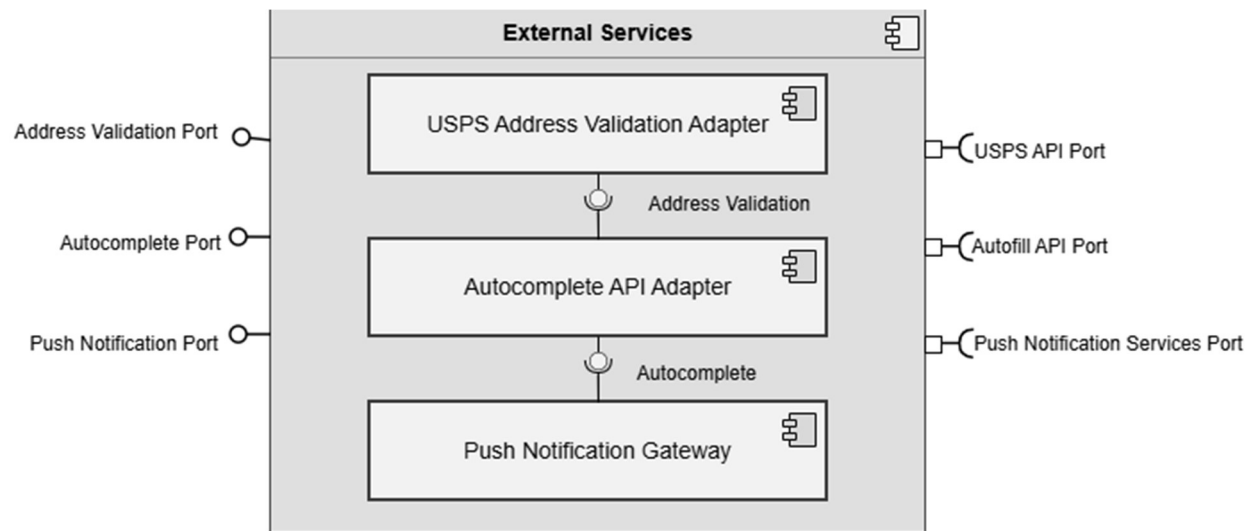
Field	Description / Purpose
	Candidate Search
	Wildcard Search
	Job Search
	Search Cache
	Company Search
	Filter Processor
	Search Engine Adapter
	Search Index / Search Engine
	Autocomplete Component
	Result Ranking
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 2.4 Job Search, 3.2 Candidate Search
Viewpoint	Component Diagram

Figure 1.10 – Communication Services (Notifications + News Feed) Detail**Design Information**

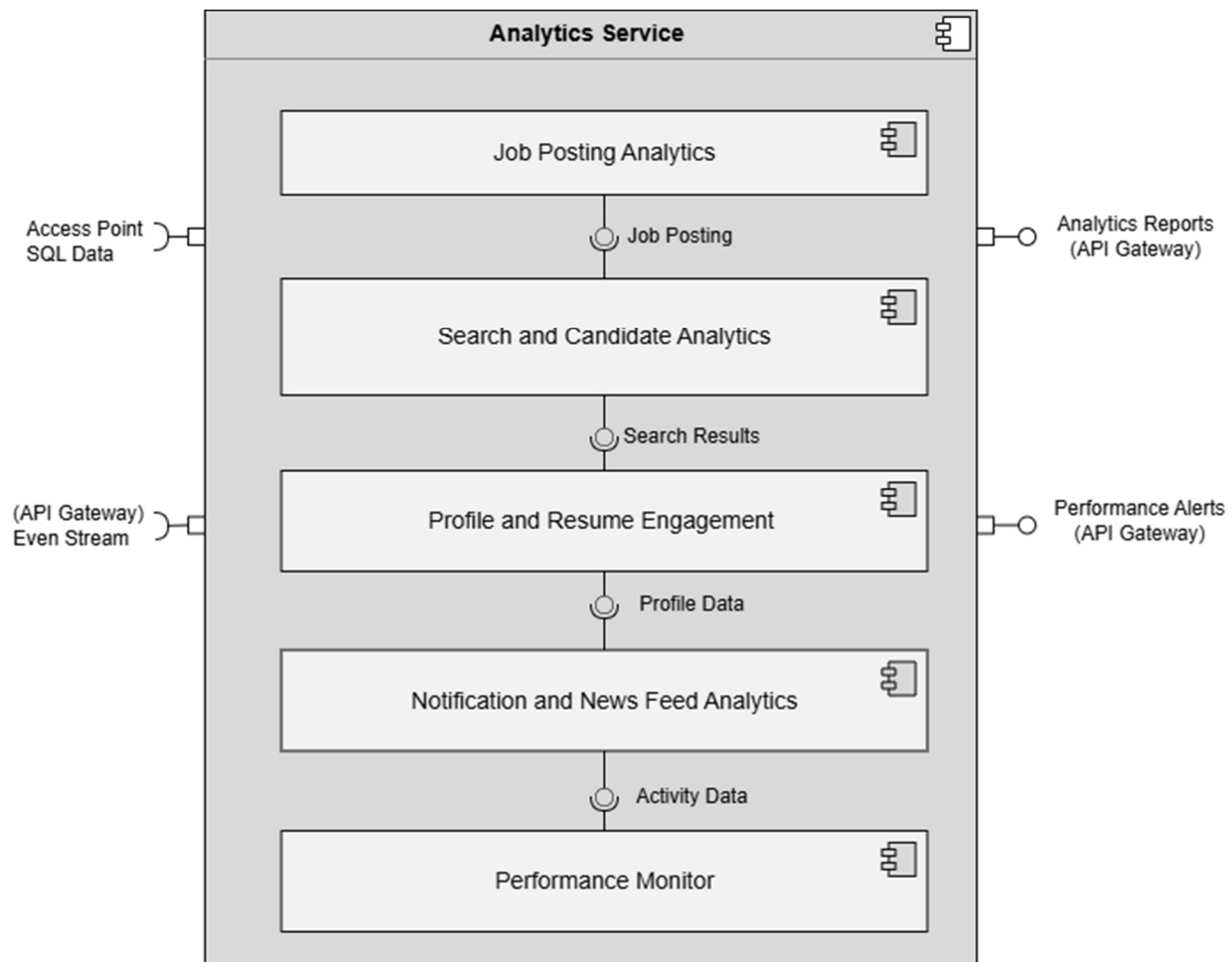
Field	Description / Purpose
View Name & Number	1.10 Communication Service
Purpose	Describe the communication service.
Description	Basic service demonstrating the communication service.
Requirements	SRS 2.2 Notifications (2.2.1–2.2.5), 2.4 News Feed (2.4.1–2.4.6), 3.2 Notifications (3.2.1–3.2.2), 3.4 News Feed (3.4.1–3.4.2)
Elements	Event Listener Notification Controller Notification Orchestrator Notification Formatter Notification Repository Recipient Resolver
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 4.1 Notifications, 4.2 News Feed
Viewpoint	Component Diagram

Figure 1.11 – Recommendation Service Detail**Design Information**

Field	Description / Purpose
View Name & Number	1.11 Recommendation Service
Purpose	Describe the recommendation service.
Description	Basic service demonstrating the recommendation service.
Requirements	Implicit in SRS 2.3 Search (2.3.7 Ranking), 2.1.3.8 (Search), 3.3 (Student Search); supports job and candidate recommendations
Elements	Job Posting Index Candidate Recommendations Skills-Based Matching Resume Listing Index Job Recommendations to Students
Navigation / References	1.4 API Gateway / Orchestrator, 1.6 Profile Management Service, 1.7 Resume Service, 1.8 Job Posting Service, 1.14 Database
Viewpoint	Component Diagram

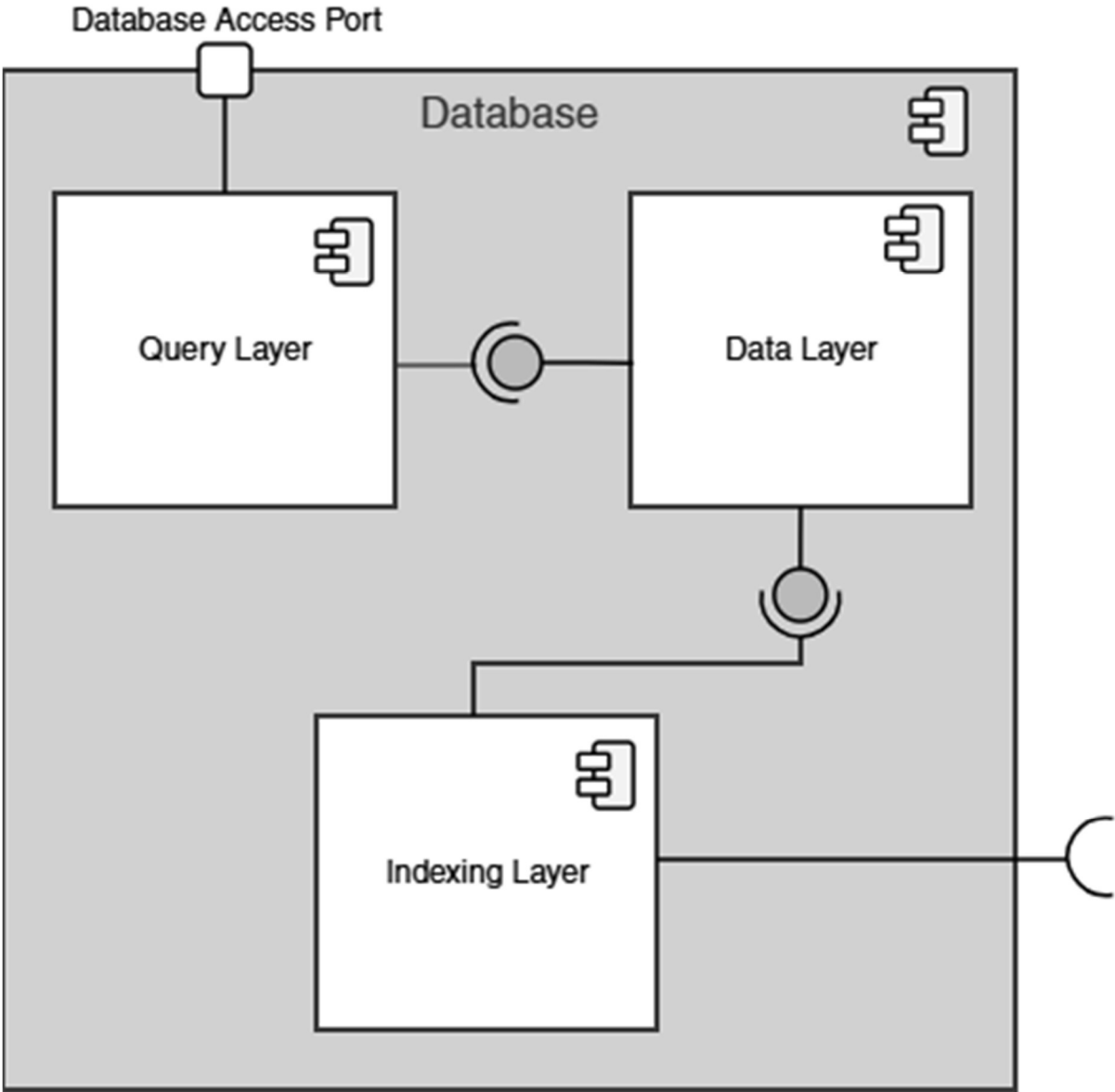
Figure 1.12 – External Services Detail**Design Information**

Field	Description / Purpose
View Name & Number	1.12 External Services
Purpose	Describe the external services.
Description	Basic service demonstrating the external services.
Requirements	SRS 2.1.3.9 (USPS address validation), 2.3.6 (Autocomplete API), 2.2.3–2.2.4 (Push notification delivery)
Elements	USPS Validation Autocomplete API Adapter Push Notifications Gateway
Navigation / References	1.4 API Gateway / Orchestrator
Viewpoint	Component Diagram

Figure 1.13 – Analytics Service Detail**Design Information**

Field	Description / Purpose
View Name & Number	1.13 Analytics Service
Purpose	Describe the analytics service.
Description	Basic service demonstrating the analytics service.
Requirements	SRS 2.1.3.6 (Company analytics), 2.5.7 (Job posting status/performance tracking)
Elements	Job Posting Analytics Search and Candidate Analytics Profile and Resume Engagement Notifications and News Feed Analytics Performance Monitor
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database
Viewpoint	Component Diagram

Figure 1.14 – Database Detail

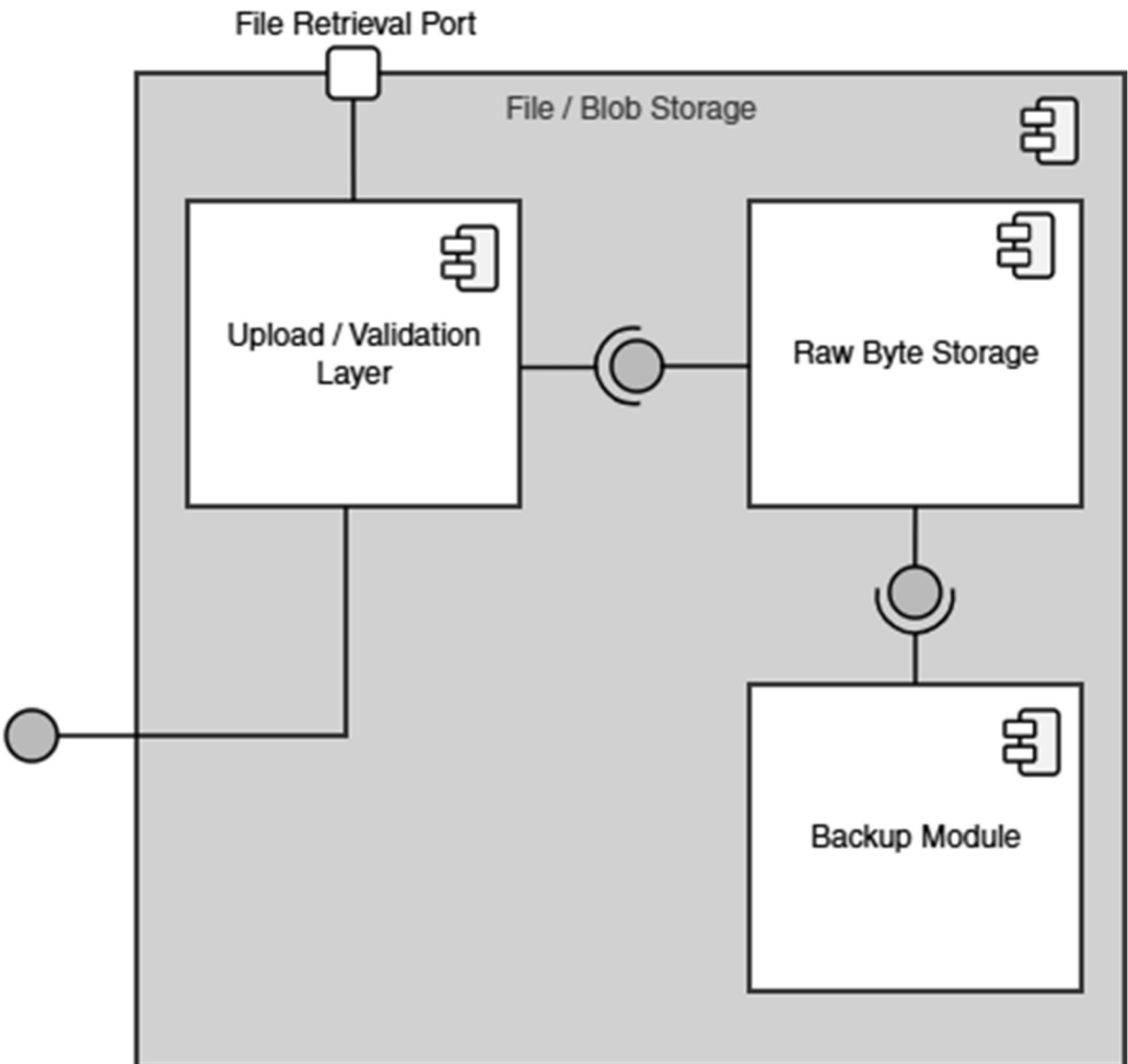


Design Information

Field	Description / Purpose
View Name & Number	1.14 Database
Purpose	Describe the database.
Description	Basic service demonstrating the database.
Requirements	All SRS sections — provides persistent storage for all services. supports 2.1, 2.2, 2.3, 2.4, 2.5, 3.1, 3.2, 3.3, 3.4, 3.5
Elements	Query Layer Data Layer Indexing Layer
Navigation /	1.5–1.13 (all service detail views)/ part of Data Storage Layer

Field	Description / Purpose
References	(Figure 1.1)
Viewpoint	Component Diagram

Figure 1.15 – File / Blob Storage Detail



Design Information

Field	Description / Purpose
View Name & Number	1.15 File / Blob Storage
Purpose	Describe the file and blob storage.
Description	Basic service demonstrating file and blob storage.
Requirements	SRS 3.5 Resume (resume file storage), 2.1.3.1 (profile image uploads), 2.1.2 (company resume access)

Field	Description / Purpose
Elements	Upload / Validation Layer
	Raw Byte Storage
	Backup Module
Navigation / References	1.14 Database; 1.6 Profile Management Service, 1.7 Resume Service; part of Data Storage Layer (Figure 1.1)
Viewpoint	Component Diagram

These sub-diagrams illustrate how each major component has been further decomposed into smaller, cohesive sub-components while preserving the explicit interfaces and ports defined at the system level. This decomposition supports modular design principles and enables independent development and maintenance of each subsystem.

3.3 Design Rationale

Discuss the rationale for selecting the architecture described in 3.1 including critical issues and trade/offs that were considered. You may discuss other architectures that were considered, provided that you explain why you didn't choose them.

4.0 DATA DESIGN

4.1 Data Description

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed and organized. List any databases or data storage items.

4.2 Data Dictionary

Alphabetically list the system entities or major data along with their types and descriptions. If you provided a functional description in Section 3.2, list all the functions and function parameters. If you provided an OO description, list the objects and its attributes, methods and method parameters.

5.0 COMPONENT DESIGN

In this section, we take a closer look at what each component does in a more systematic way. If you gave a functional description in section 3.2, provide a summary of your algorithm for each function listed in 3.2 in procedural description language (PDL) or pseudocode. If you gave an OO description, summarize each object member function for all the objects listed in 3.2 in PDL or pseudocode. Describe any local data when necessary.

6.0 HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The following diagrams provide an approximation of the central UI elements that will allow the users to utilize the various services provided, including descriptions of their functionality and purposes as well as site navigation details.

6.2 Screen Images

1.1 Login / Logout

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Login/Logout Email* NotMyLover@gmail.com... Password* Enter Password... Login New User? Register here	

Mobile View

 CareerBound	 
Search Bar	
Login/Logout	
Email*	
NotMyLover@gmail.com...	
Password*	
Enter Password...	
Login	
New User? Register here	

Design Information

Field	Description / Purpose
View Name & Number	1.1 Login / Logout
Purpose / Design Concern	Provides a way for students and companies to log into their profiles.
Description	A website page which displays a log in field for the user to input their email and password to authenticate themselves.
Requirements	Authentication requirements supporting all protected system functionality under Sections 2.x and 3.x
Elements	Email Field
	Password Field
	Login Button
	Invalid Login Information Field
	Registration Redirect Button
Inputs	Email and Password fields, Login button click, Register link click, Forgot Password link click.
Outputs	Redirect to Dashboard (2.1 for students, 3.1 for companies) if user log in was successful. Displays an error message if invalid credentials were provided.
Navigation / References	Student Dashboard 2.1, Company Dashboard 3.1, Registration 1.2
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile).

1.2 Registration

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Registration Back to Login Student / Employer* Employer ▼ First Name* Billy... Last Name* Jean... Email* NotMyLover@gmail.com... Password* Enter Password... ☒ 8 characters long ☒ Includes capital letter ☐ Includes special character Confirm Password* Enter Password... Submit	

Mobile View

 CareerBound  

Search Bar

Registration

Student / Employer*

Employer ▾

First Name*

Billy..

Last Name*

Jean...

Email*

NotMyLover@gmail.com...

Password*

Enter Password...

☒ 8 characters long

☒ Includes capital letter

☐ Includes special character

Confirm Password*

Confirm Password...

Submit

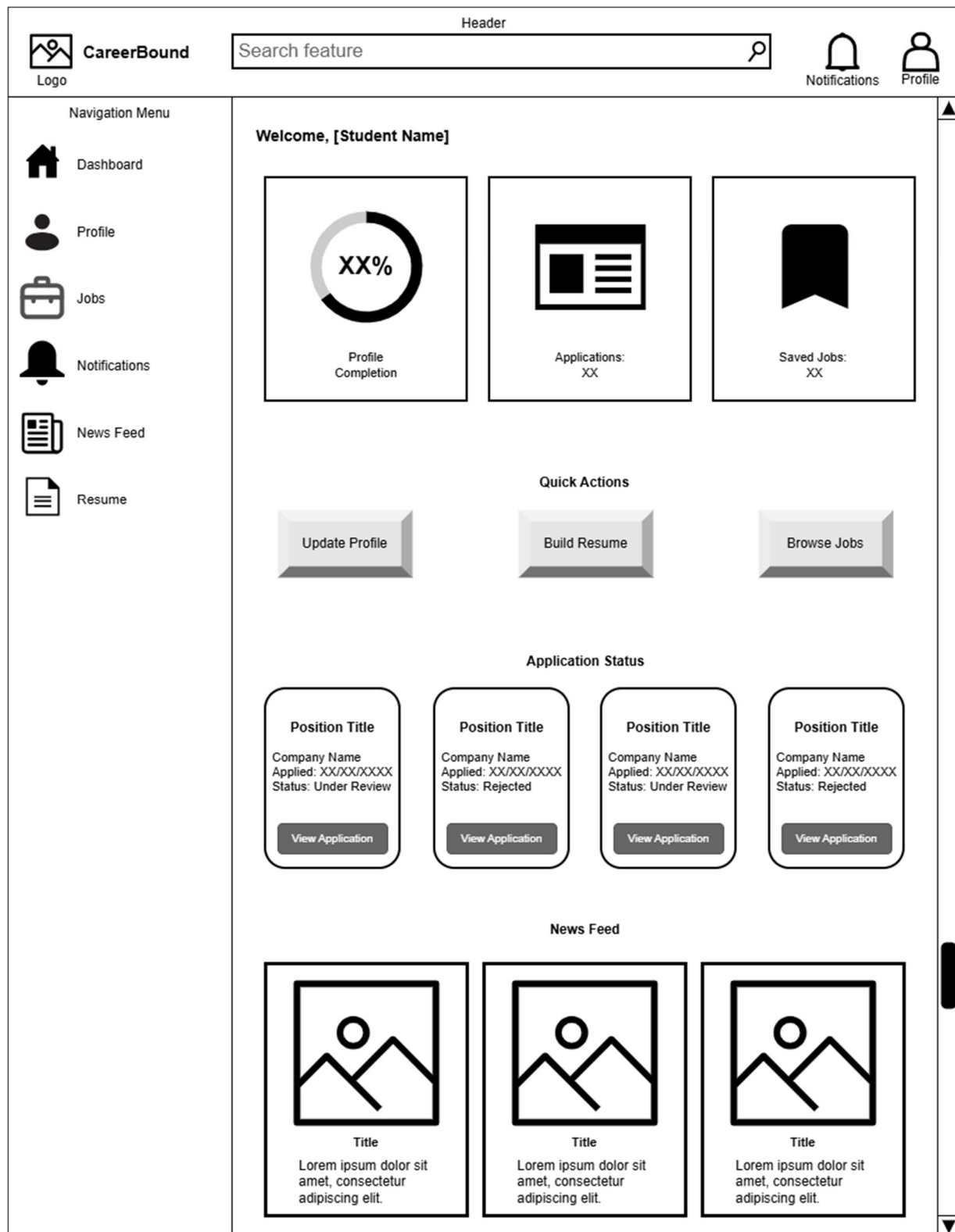
Back to Login

Design Information

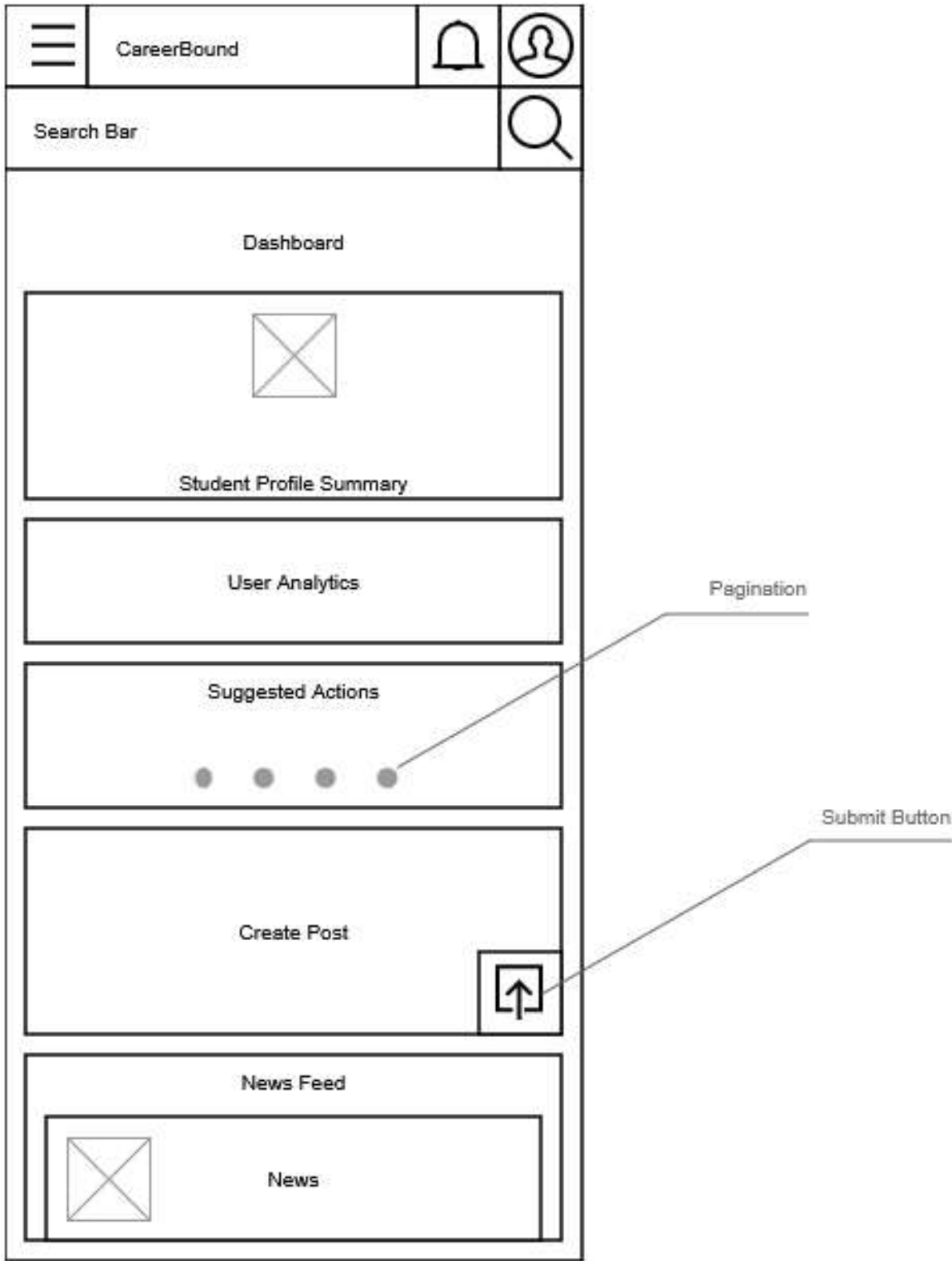
Field	Description / Purpose
View Name & Number	1.2 Registration
Purpose / Design Concern	Facilitates user account creation and role selection for new users.
Description	Displays a registration form allowing a user to select their role and establish account credentials.
Requirements	SRS 2.1 Company Profile Page, 3.1 Student User Profile
Elements	Back to Login button
	Student / Employer role selection dropdown
	Text fields
	Password validation checklist
	Submit button
Inputs	Role selection, First Name string, Last Name string, Email string, Password string, Confirm Password string, Submit action.
Outputs	Visual feedback on password requirement validation (status icons for "8 characters long", "Includes capital letter", and "Includes special character").
Navigation / References	1.1 Login / Logout (accessed via "Back to Login" button). 2.1 Student Dashboard or 3.1 Company Dashboard upon successful submission.
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile).

2.1 Student Dashboard

Desktop View



Mobile View



Design Information

Field	Description / Purpose
View Name & Number	2.1 – Student Dashboard
Purpose / Design Concern	Provides students with a centralized dashboard for managing career development activities, tracking applications, accessing important tools, and viewing personalized updates within the Career Bound system.
Description	Displays a summarized student profile; profile progress (desktop) provides analytics such as profile views, profile completion, matching jobs, post impressions, application summaries; application tracking cards (desktop); recommends actions (mobile) to take such as improve profile, create resume, apply for a job; provides a shortcut to post creation; and displays social media-style news feed containing career-related updates and announcements.
Requirements	3.1 Student User Profile, 3.2 Notifications, 3.3 Search Features, 3.4 News Feed, 3.5 Resume
Elements	Student Profile Summary Card (mobile)
	Student User Analytics Card (mobile)
	Suggested Actions Card (mobile)
	Quick Create Post Card (mobile)
	News feed List
	News feed Cards
	Submit Button
	Navigation menu (Sidebar for desktop, toggle for mobile)
	Header
	Search Feature
	Applications Summary Card (desktop)
	Saved Jobs Summary Card (desktop)
	Applications Status Cards (desktop)
	Notifications toggle button
	Profile Page hyperlink button
	Pagination
Inputs	Student Post, Click, Scroll, Search queries, navigation selections, notification interactions, profile selections, quick action button selections, application card selections, news feed interactions
Outputs	Profile completion percentage, application status information, saved job totals, recommended actions, job application details, career updates, company announcements, News feed content, New Student Post, Navigation redirects, Navigate to next action card
Navigation / References	Authentication, Student Profile, Resume Builder, Job Search, Job Details, Notifications, News Feed, Saved Jobs, Job Applications
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

2.2 Student Profile

Desktop View

CareerBound	Search Bar	 
Dashboard Profile Jobs Resume Notifications News Feed	Section Title	
	Profile Picture Name: Email: Phone: Bachelors Degree: Skills: Linkedin Profile	
	About Me	
	Edit Info Save	
	Delete Profile	

Mobile View

	CareerBound	 
Search Bar [Filter]		
Section Title		
Profile Picture		
Name: Email: Phone: Bachelors Degree: Skills: Linkedin Profile		
About Me		
Edit Info Save Delete Profile		

Design Information

Field	Description / Purpose
View Name & Number	2.2 – Student Profile
Purpose / Design Concern	Allows students to view and manage their profile information.
Description	Displays student information such as name, contact information, program, skills, and resume-related details.
Requirements	3.1 Student User Profile (3.1.1–3.1.3.3)
Elements	Logo
	Search bar
	Menu/Profile
	Left Navigation
	Section Title
	Profile Picture
	User Information
	About Me
	Save/Delete buttons
Inputs	Updated student profile information, Upload CV
Outputs	Displayed student profile information
Navigation / References	Dashboard, Profile, Jobs, Resume, Notifications, News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

2.3 Resume Builder

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Resume Editor Tools	
	Resume Viewer	

Mobile view

CareerBound

Search Bar

Resume 1
User Tag: LoremIpsilum123

Resume 2
User Tag: LoremIpsilum123

+

CareerBound

Search Bar

Edit Resume

Personal Info

Education

Experiences

Skills

Full Name *

Email *

Phone Number

Submit

Design Information

Field	Description / Purpose
View Name & Number	2.3 Resume Builder
Purpose / Design Concern	Provides a space that allows students to view and edit their resume that is to be used.
Description	Displays student's resume and gives tools to make and save changes to the resume.
Requirements	SRS 3.5 Resume (3.5.1–3.5.5.2), 3.5.4 User Input, 3.5.5 Input Submission
Elements	Logo
	Search Bar
	Profile/Menu
	Dashboard
	Profile
	Jobs
	Resume
	Notifications
	News Feed
	Resume Editor Tools: Edit Resume Section Tabs/ Add New Resume Button/ Helpful info buttons/ Submit Resume Button/ Asterisks denoting Required Data Entry
	Resume Viewer: Resume List
Inputs	Resume Edits, Search terms, Resume data entries, Wildcard search
Outputs	Resume reflecting saved changes and recent changes made, Matching resume data terms, resume list search results, new resume data entries
Navigation / References	Profile/Menu, Dashboard, Profile, Jobs, Resume, Notifications, News Feed, Resume List, Edit Resume Screen
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

2.4 Job Search

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	[Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris [Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris [Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris	[Active Search Filters] X [Filter 1] X [Filter 2]

Mobile View

 CareerBound



Search Bar

[Filter]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

Design Information

Field	Description / Purpose
View Name & Number	2.4 – Job Search
Purpose / Design Concern	Allows students to search for jobs and companies.
Description	Displays searchable job listings with filters and detailed results.
Requirements	SRS 2.3 Search Features, 3.3 Student Search Features
Elements	Search bar
	Filters
	Results list
	Save/apply buttons
Inputs	Search terms, filter selections, wildcard search
Outputs	Matching job postings and company search results
Navigation / References	Job Details Screen, Company Profile, Student Dashboard
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

2.5 Job Details

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard	[Job Title] [Company Name] [City, State] [Full-time] [Pay Rate] Apply Button	
Profile		
Jobs		
Resume		
Notifications		
News Feed		
	About the Role Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Responsibilities • Lorem ipsum dolor • Sit amet, consectetur adipiscing Nice to Have • Lorem ipsum dolor • Sit amet, consectetur adipiscing <td> About [Company Name] [Company Website] [Company Size] [Industry] [Headquarters] Job Details [Job ID] [Department Name] [Experience Level] [In-Person/Remote/Hybrid] Required Skills [Skill 1] [Skill 2] [Skill 3] [Skill 4] </td>	About [Company Name] [Company Website] [Company Size] [Industry] [Headquarters] Job Details [Job ID] [Department Name] [Experience Level] [In-Person/Remote/Hybrid] Required Skills [Skill 1] [Skill 2] [Skill 3] [Skill 4]

Mobile View

 CareerBound



Search Bar

[Filter]

 Back to Search Results

[Job Title]
[Company Name]
[City, State]
[Employment Type]
[Pay Rate]

Apply Button

About the Role

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum

Responsibilities

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod
- tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

Job Details

[Job ID]
[Department Name]
[Experience Level]
[In-Person/Remote/Hybrid]

Nice to Have

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod
- tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

About [Company Name]
[Company Website]
[Company Size]
[Industry]
[Headquarters]

Required Skills

[Skill 1]
[Skill 2]
[Skill 3]
...

46

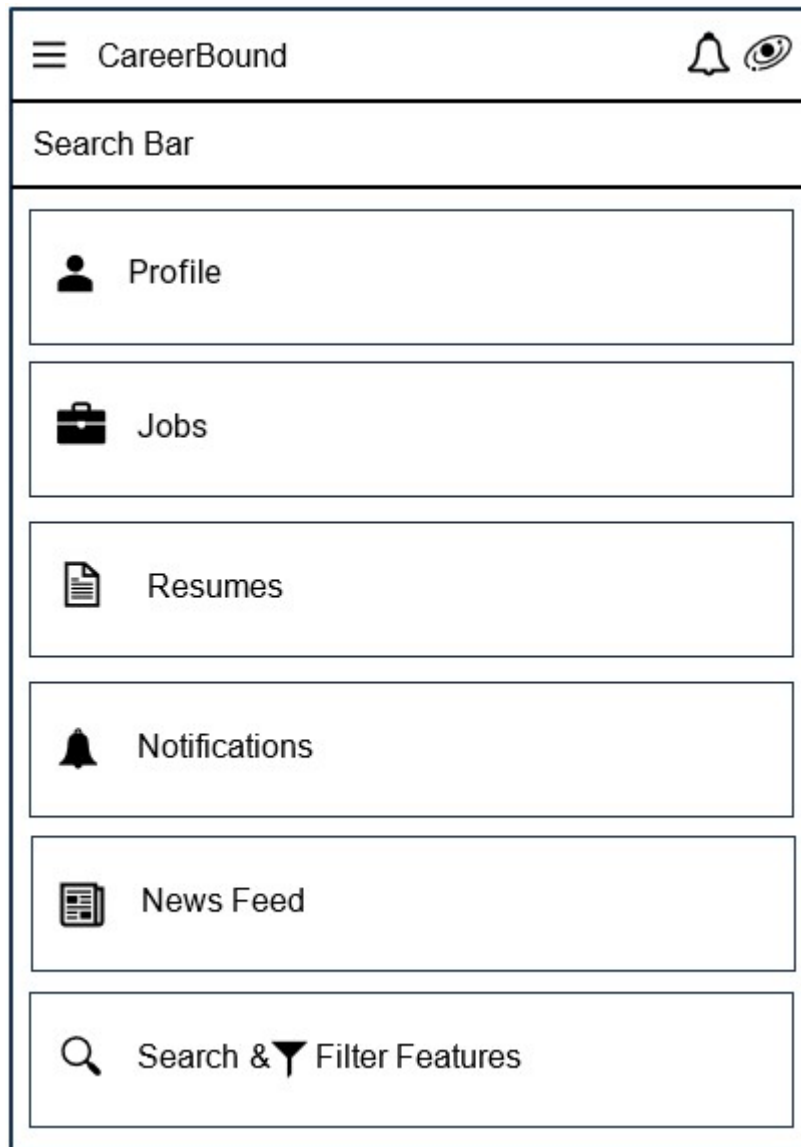
Design Information

Field	Description / Purpose
View Name & Number	2.5 – Job Details
Purpose / Design Concern	Allows students to get more information for Job Postings.
Description	Displays all information relevant to a given Job Postings.
Requirements	SRS 2.5.2, SRS 2.5.7, SRS 2.5.9
Elements	Back Button
	Job Data
Inputs	Tap/Click on Apply Button
Outputs	Application Submission Confirmation, Updated Application Status
Navigation / References	Job Search Screen, Company Profile, Student Dashboard
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

3.1 Company Dashboard

Desktop View

CareerBound [Company Logo]	🔍 Search Bar	🔔 Notifications	Company Profile																				
Navigation	3.1 Company Dashboard																						
Dashboard	Active Job Posting 08	Applications 42	Candidate Search 16																				
Profile			Notifications 05																				
Jobs	Quick Actions Create Job Posting Candidate Search View Resumes	Resumes <table><tr><th>Candidate</th><th>Job Posting</th><th>Status</th><th>Date</th></tr><tr><td>A. Rivera</td><td>Platform Engineer</td><td>Under Review</td><td>05/15/26</td></tr><tr><td>T. Nguyen</td><td>Data Analyst</td><td>New</td><td>05/14/26</td></tr><tr><td>A. Patel</td><td>UX Researcher</td><td>Interview</td><td>05/13/26</td></tr><tr><td>J. Kim</td><td>Product Manager</td><td>Complete</td><td>05/15/26</td></tr></table>		Candidate	Job Posting	Status	Date	A. Rivera	Platform Engineer	Under Review	05/15/26	T. Nguyen	Data Analyst	New	05/14/26	A. Patel	UX Researcher	Interview	05/13/26	J. Kim	Product Manager	Complete	05/15/26
Candidate	Job Posting	Status	Date																				
A. Rivera	Platform Engineer	Under Review	05/15/26																				
T. Nguyen	Data Analyst	New	05/14/26																				
A. Patel	UX Researcher	Interview	05/13/26																				
J. Kim	Product Manager	Complete	05/15/26																				
Resume																							
Notifications																							
News Feed	Job Posting Status Active 08 Under Review 03 Expiring Soon 02	News Feed CareerBound Update New company insights available for active Job Posting and Applications. Hiring Trends Candidate Search activity increased for software and analytics roles this week.																					

Mobile View**Design Information**

Field	Description / Purpose
View Name & Number	3.1 – Company Dashboard
Purpose / Design Concern	Company dashboard to help a company manage recruiting activity from desktop and mobile views.
Description	Provides a desktop and mobile view where company users can review job posting activity, applications, candidate search access, notifications, and company-related updates.
Requirements	Software Requirements Specification (SRS) for Career Bound: 2.1 Company Profile Page, 2.2 Notifications, 2.3 Search Features, 2.4 News Feed, 2.5 Job Postings

Field	Description / Purpose
	Reference: https://content.byui.edu/file/968cdd33-09eb-4430-99a4-131d25e138d4/1/Teach/SRS/430.SRS.CareerBound.pdf
Elements	Navigation menu: provides a company with a list of options to select from.
	Search bar: provides a company the ability to search.
	Company Profile area: provides a company with access to all company profile data.
	Notifications area: provides a company with access to company notifications.
	Candidate Search feature: provides a company the ability to search or filter job candidates.
	Job Posting area: provides a company access to job postings.
	Resume section: provides a company access to a list of candidates who have submitted job applications.
	News Feed section: provides a company access to news feed options
	Login / Logout: clicking the company logo provides a company with the ability to login or logout.
Inputs	Search query text, clicks on navigation items, clicks on quick action buttons, and selections related to applications or job postings.
Outputs	Company dashboard metrics, application status information, job posting status information, notifications, and news feed updates.
Navigation / References	1.1 Login / Logout, 3.2 Candidate Search, 3.3 Job Posting Creation, 4.1 Notifications, 4.2 News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

3.2 Candidate Search

Desktop View

Logo	Search Bar	Profile/Menu	
Dashboard Profile Jobs Resume Notifications News Feed	 Search here Advanced Search Results for "Lorem Ipsum" 5 Results John Doe Senior  Jane Doe Junior  Joe Smith Senior  Sue Smith Junior  Mike Jackson Senior 		
	Filters		
	Education		
	Major		
	Technical Skills		

Mobile View

 CareerBound
 


Search Bar

 Search here
 

Results for "Lorem Ipsum"

5 Results
 Filters

John Doe

Education

☒
☐
☐

Major

☒
☐
☐

Technical Skills

☒
☒
☐

Jane Doe

Senior

☐
☐
☐

☐
☐
☐

☐
☐
☐

Joe Smith

Senior

☐
☐
☐

Design Information

Field	Description / Purpose
View Name & Number	3.2 Candidate Search
Purpose / Design Concern	Allows company recruiters a way to find potential job seekers.
Description	A specialized search engine for the user to find candidates based on desired filters. Designed for both desktop users and mobile users.
Requirements	2.3.1, 2.3.2, 2.3.4, 2.3.8
Elements	Search Bar Site Navigation Search Results Grid Filter Categories Filter Checkboxes Search Result Cards

Field	Description / Purpose
	Advanced Search Toggle
	Candidate Name
	Candidate Profile Picture
	Candidate Year
	Filter Scroll Bar
Inputs	User will click on the search bar and enter a search item; user can choose a filter to toggle; the user can choose a candidate to open a new page.
Outputs	Candidate profile cards display candidate information; candidate profiles can be filtered.
Navigation / References	3.1 Company Dashboard, 3.3 Job Posting Creation
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

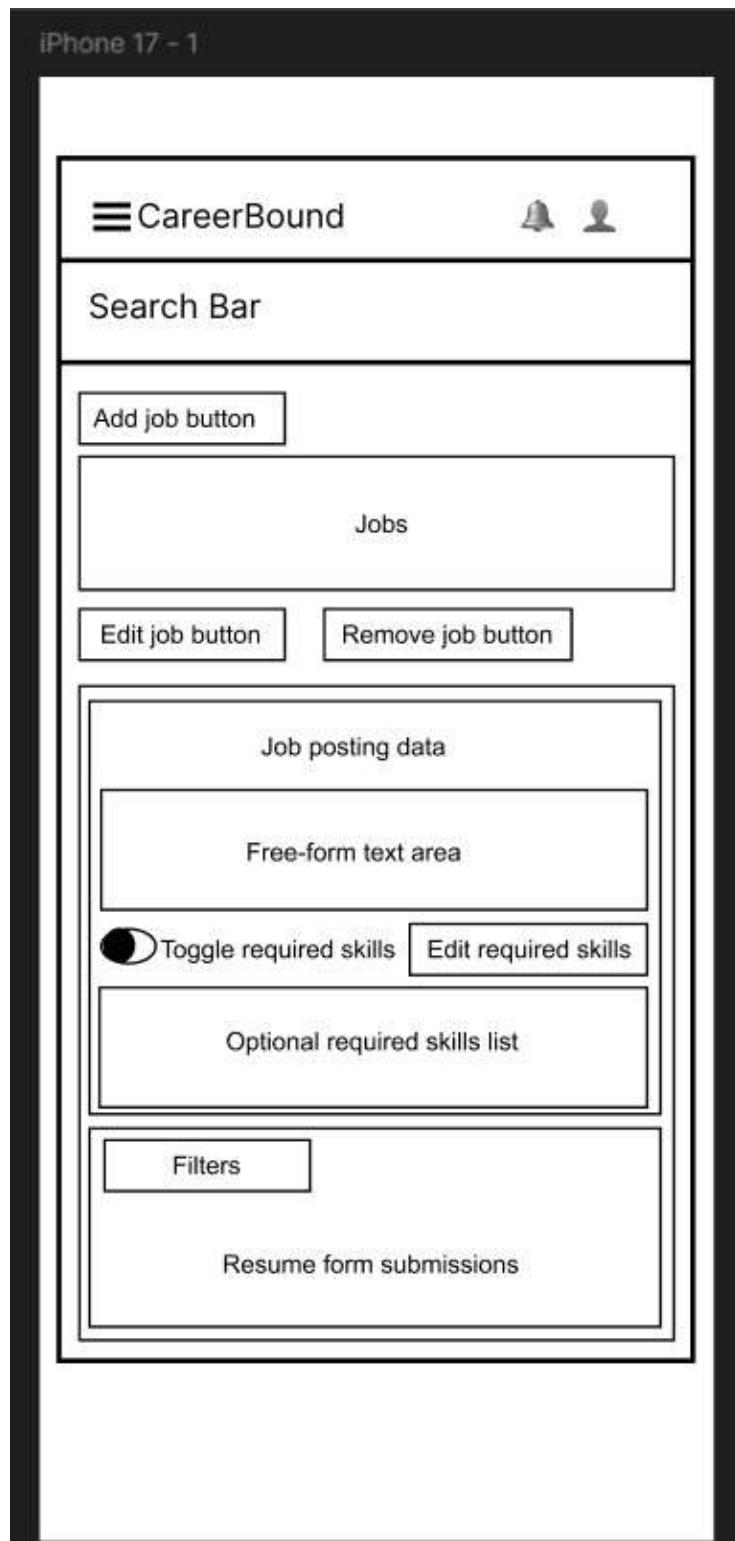
3.3 Job Posting Creation

Desktop View

Desktop - 1

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Add job button Edit job button Remove job button Jobs	Job posting data Job posting data Free-form text area ☒ Toggle required skills posting Edit required skills Optional required skills list Filters Resume form submissions

Mobile View



Design Information

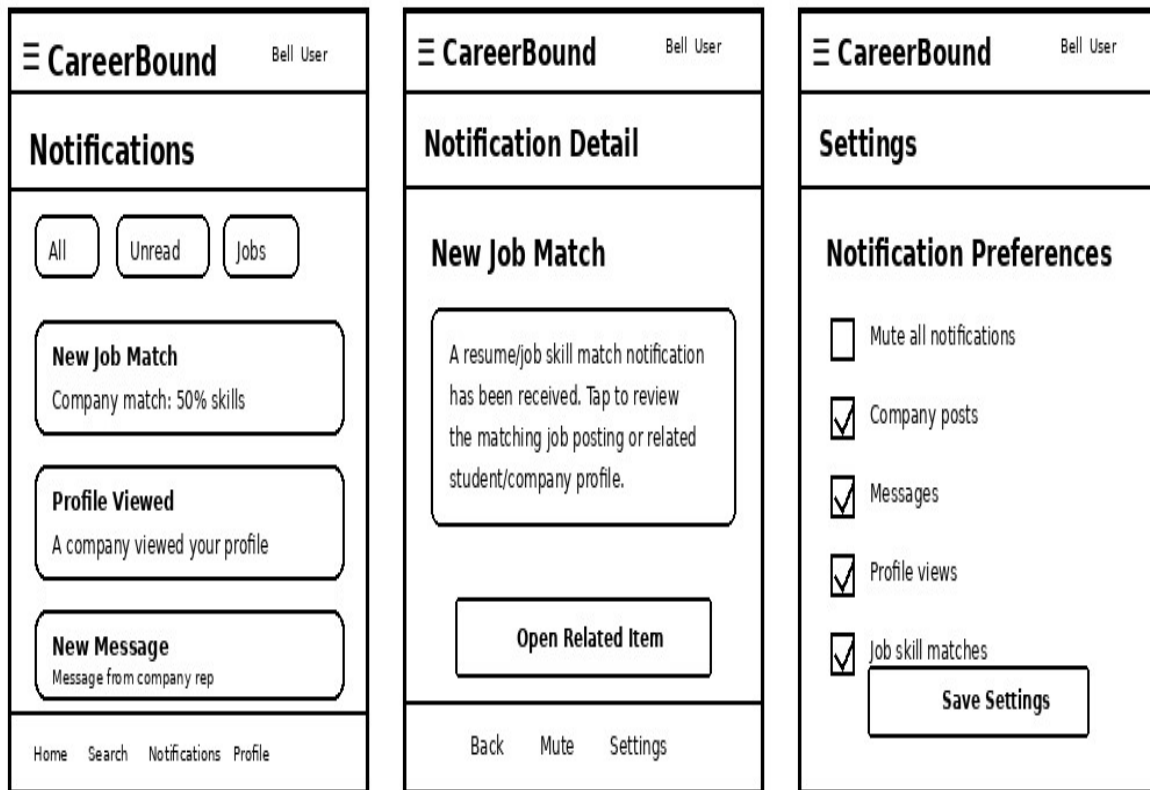
Field	Description / Purpose
View Name & Number	3.3 Job Posting Creation
Purpose / Design Concern	Allows company users to create, edit, and remove job postings, manage required skills, and review resume submissions in a centralized Jobs view.
Description	Displays a company-facing Jobs screen with job posting controls, a job posting data panel, a free-form text area, a required skills toggle, an editable required-skills list, a filters section, and a resume form submissions area in both desktop and mobile wireframe layouts.
Requirements	SRS 2.5 Job Postings (2.5.1--2.5.7)
Elements	Header with logo, search bar, and profile/menu
	Navigation menu with Dashboard, Profile, Jobs, Resume, Notifications, and News Feed
	Add Job Button
	Edit Job Button
	Remove Job Button
	Jobs Panel
	Job Posting data section
	Free-form text area
	Required skills toggle
	Edit Required Skills button
	Optional required skills list
	Filters section
	Resume form submission panel
Inputs	Company user selects Add Job, Edit Job, or Remove Job; enters or edits job posting text; toggles required skills on or off; edits the required skills list; applies filters; reviews submitted resumes.
Outputs	Displayed job posting information, updated required skills settings, filtered submission results, and visible resume form submissions for review.
Navigation / References	Related views include Company Dashboard (3.1), Candidate Search (3.2)
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

4.1 Notifications

Desktop View

Logo	Search Bar	Profile/Menu
Dashboard	Notification ☒ Push Notifications News Feed New Post by companies you are following Receive notification from companies that you are following Personal message from company Company Representative Views profile ☐☐☐☐☐☐	
Profile		
Jobs		
Resume		
Notifications	Notification Filters ☒ Create Filter Example Filter ✕	
News Feed		

Mobile View



Design Information

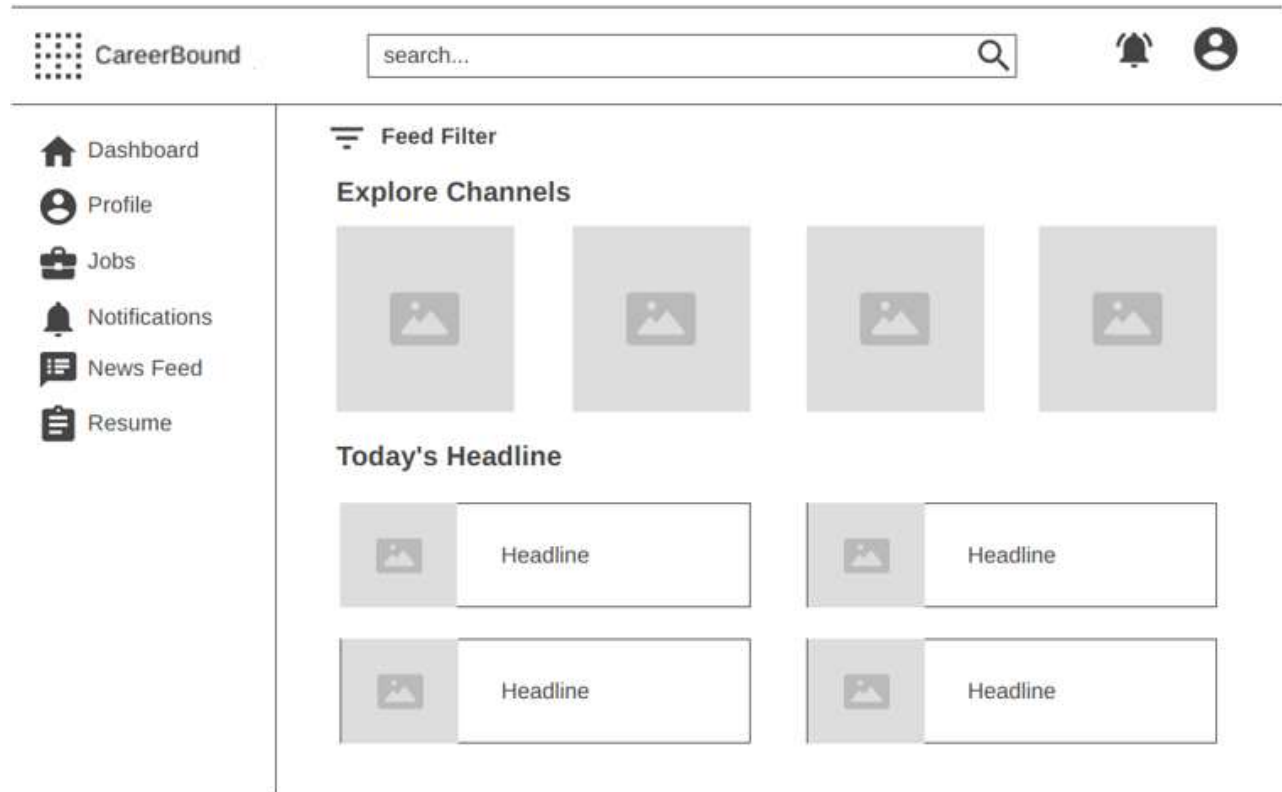
Field	Description / Purpose
View Name & Number	4.1 Notifications Screen
Purpose / Design Concern	The Notification Screen exist to allow the user to customize what notifications they receive from Career Bound. This is for the End User and satisfies section 2.2 / 3.2 in the SRS.CareerBound.docx
Description	The Notification Screen will allow the user to customize notifications about their News Feed, Company Viewing their profile, Push Notifications, New posts by company they follow, notifications from company they follow, personal messages, and allow the user to mute all notifications. Uses the filter component to create a list of companies to receive notifications from. Shows mobile notification screens for student users and company representatives, including the notification list, notification detail screen, push notification handling, and

Field	Description / Purpose
	notification preferences.
Requirements	1.2.1 Application Scope; 2.2 Notifications; 3.2.1. There shall be a notification push sent to the student user instantly after a notification trigger. 3.2.2. The student user shall receive a notification when a company they are following makes a post or sends out a notification to its followers. 3.2.3. The student user shall receive a notification when a company sends them a personal message. 3.2.4. Notifications may be pushed to a student user's phone outside of the system 3.2.5. Student users may receive a notification when a company representative views his/her profile. 3.2.6. All users may choose to mute notifications 3.2.7. There shall be settings allowing the student user to add, alter and remove filters for notifications.
Elements	Search Bar App navigation Toggle Switch Create Filter Button Edit Pen Icon Delete X Icon Notification Card: Notification Screen; Notification List; Notification Detail View; Push Notification Handler; Notification Preferences; Mute Notification Control; Notification Filter Card: Filter Settings; Profile/Menu API Client Adapter / REST/HTTP API Port
Inputs	Toggle Switch, Click Button, Click Text, Filter Companies, Search, Selected notification, filter settings, mute setting changes, push notification trigger, and mobile navigation actions
Outputs	Notifications Settings, Filtered Company List, Notification list, notification detail information, filtered notification results, push notification display, and updated notification settings
Navigation / References	Filter Create Screen, Dashboard Screen, Profile Screen, Jobs Screen, Resume Screen, News Feed Screen, Mobile Client,

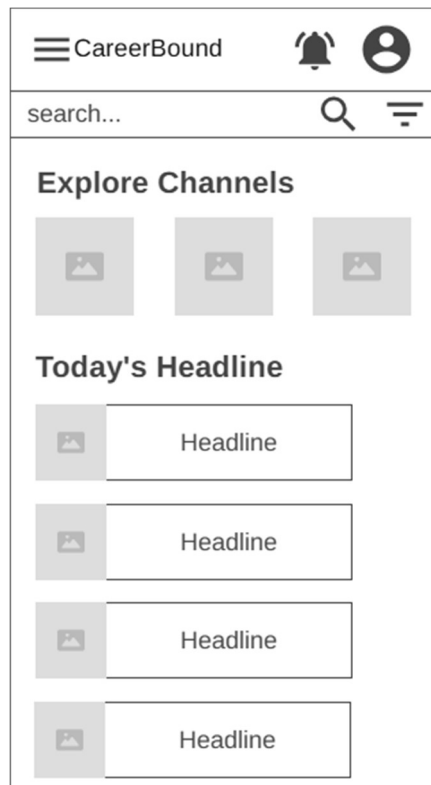
Field	Description / Purpose
	Notification Screen, Notification Detail View, Notification Preferences, API Gateway / Orchestrator
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

4.2 News Feed

Desktop View



Mobile View



Design Information

Field	Description/Purpose
View Name & Number	4.2 – News Feed
Purpose / Design Concern	Allows students to filter and view their news feed.
Description	Displays relevant news for students, which can be filtered by different tags to view specific news reports.
Requirements	2.4 News Feed (2.4.1–2.4.6), 3.4 News Feed (3.4.1–3.4.2)
Elements	Logo Notifications Search Bar Menu/Profile Feed Filter Explore Channels Headlines
Inputs	Current news, current location
Outputs	Updated news feed, current location displayed
Navigation / References	Dashboard, Profile, Jobs, Resume, Notifications, News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

6.3 Screen Objects and Actions

A discussion of screen objects and actions associated with those objects.

7.0 REQUIREMENTS MATRIX

The following table provides a cross-reference that traces each major component in the system architecture to the specific SRS requirements it fulfills. Below this SRS Requirements Mapping Table you will find Requirements Traceability Matrices for quick visual SRS Requirements tracing.

SRS Requirements Mapping Table

Component	SRS Sections / Requirements It Directly Supports
Client Applications (Parent)	• 1.2 Application Scope (web and mobile interfaces that share the same features) • Overall client-side architecture, shared UI patterns, role-based views, and coordination between web and mobile clients
Web Client	• Primary web-based user interface (browser UI) • Implements all web UI elements for Company and Student features (sections 2.x and 3.x) • Profile pages, search bar, forms, news feed display, notifications display, resume form, job browsing, etc. (1.2.1)
Mobile Client	• Mobile application user interface • Implements the same features and characteristics as the Web Client (per SRS 1.2.1) • Mobile-optimized UI for all features in sections 2.x and 3.x (profiles, search, news feed, notifications, resume form, job browsing, etc.)
API Gateway / Orchestrator	• Cross-cutting orchestration for all client–service communication (implicit in 1.2 and all functional requirements in 2.x / 3.x)
Authentication Service	• Foundational requirement for all logged-in features (profiles, following, notifications, job postings, resume access, etc.) • Required for every section under 2.x and 3.x
Profile Management Service	• 2.1 Company Profile Page (2.1.1–2.1.3.10) • 3.1 Student User Profile (3.1.1–3.1.3.3)
Resume Service	• 3.5 Resume (3.5.1–3.5.5.2) • Referenced in 2.1.2, 2.2.5, 2.3.3, 2.3.5, 2.5.6, 3.1.3.1–3.1.3.3
Job Posting Service	• 2.5 Job Postings (2.5.1–2.5.7) • Referenced in 3.3.1, 3.4.1, 2.2.5
Search Service	• 2.3 Search Features (2.3.1–2.3.8) • 3.3 Search Features (3.3.1–3.3.2)

Component	SRS Sections / Requirements It Directly Supports
Communication Services (Notifications + News Feed)	• 2.2 Notifications (2.2.1–2.2.5) • 2.4 News Feed (2.4.1–2.4.6) • 3.2 Notifications (3.2.1–3.2.7) • 3.4 News Feed (3.4.1–3.4.2)
Recommendation Service	• Job and candidate recommendations based on profiles, resumes, and job postings • Supports 2.3 Search Features, 2.5 Job Postings, 3.3 Search Features, and 3.1/3.5 profile/resume data
External Services	• External integrations (USPS address validation in 2.1.3.9, autocomplete API in 2.3.6, push notification services in 2.2, etc.) • Any third-party APIs referenced throughout the SRS
Analytics Service	• Provides reporting and analytics on job posting performance (views, applications, 2.5), candidate search effectiveness (filters, results, 2.3), profile and resume engagement (2.1, 3.1, 3.5), and notification / news feed activity (2.2, 2.4, 3.2, 3.4) • Monitors key performance metrics (e.g., search results returned within 3 seconds per 2.3.5) • Supports data-driven platform improvements, usage insights, and overall system monitoring
Data Storage Layer (Parent)	• Provides the single unified SQL / Data Access Port used by every service in the system • Acts as the central enclosing component that coordinates Database and File/Blob Storage • Manages system-wide data concerns (transactions, consistency, backups, scalability, reliability) • 4. Hardware requirements (implicit backend storage, servers, infrastructure)
Database	• All persistent data storage for profiles, resumes, job postings, notifications, news feed, authentication, etc. • Referenced in every “Data X.X.X” entry throughout the SRS
File / Blob Storage	• Storage of uploaded resumes, profile images, attachments, and other binary files • Supports 3.5 Resume submission and any file-related requirements

The following Requirements Traceability Matrices map the requirements defined within the Software Requirements Specification (SRS) to the CareerBound system architecture components responsible for supporting, implementing, or interacting with those requirements.

These matrices are intended to:

- verify that all documented requirements are addressed within the system design,
- demonstrate traceability between requirements and architectural components,

- support validation and quality assurance activities, and
- ensure consistency between the SRS and the Software Design Description (SDD).

An “X” within a matrix cell indicates that the corresponding component directly supports, implements, manages, or participates in fulfilling the specified requirement.

The matrices are organized by major SRS requirement sections for improved readability and maintainability.

7.1 Company Profile Requirements Matrix (SRS 2.1)

Component	2.1.1 Profile	2.1.2 Resume	2.1.3.1 Contact	2.1.3.2 Location	2.1.3.3 Skills	2.1.3.4 Following	2.1.3.5 Posting	2.1.3.6 Analytics	2.1.3.7 Messaging	2.1.3.8 Search	2.1.3.9 Validation	2.1.3.10 Security
Web Client	X	X	X	X	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X	X	X	X	X
Profile Management Service	X	X	X	X	X	X	X	X	X	X	X	X
Resume Service		X			X			X				
Job Posting Service				X		X						
Search Service		X				X				X	X	
Communication Services			X			X			X			
Recommendation Service		X				X				X		
External Services											X	

Component	2.1.1 Profile	2.1.2 Resume	2.1.3.1 Contact	2.1.3.2 Location	2.1.3.3 Skills	2.1.3.4 Following	2.1.3.5 Posting	2.1.3.6 Analytics	2.1.3.7 Messaging	2.1.3.8 Search	2.1.3.9 Validation	2.1.3.10 Security
Analytics Service		X		X	X	X			X			
Database	X	X	X	X	X	X	X	X	X	X	X	X
File / Blob Storage		X				X						

7.2 Notification Requirements Matrix (SRS 2.2)

Component	2.2.1 Alerts	2.2.2 Updates	2.2.3 Push	2.2.4 Delivery	2.2.5 Preferences
Web Client	X	X	X	X	X
Mobile Client	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X
Authentication Service	X	X	X	X	X
Profile Management Service					
Resume Service					
Job Posting Service					

Component	2.2.1 Alerts	2.2.2 Updates	2.2.3 Push	2.2.4 Delivery	2.2.5 Preferences
Search Service					
Communication Services	X	X	X	X	X
Recommendation Service					
External Services			X	X	
Analytics Service	X	X	X	X	X
Database	X	X	X	X	X
File / Blob Storage					

7.3 Search Requirements Matrix (SRS 2.3)

Component	2.3.1 Candidate	2.3.2 Company	2.3.3 Filters	2.3.4 Wildcards	2.3.5 Performance	2.3.6 Autocomplete	2.3.7 Ranking	2.3.8 Resume
Web Client	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X

Component	2.3.1 Candidate	2.3.2 Company	2.3.3 Filters	2.3.4 Wildcards	2.3.5 Performance	2.3.6 Autocomplete	2.3.7 Ranking	2.3.8 Resume
Authentication Service	X	X	X	X	X	X	X	X
Profile Management Service		X	X					
Resume Service			X					
Job Posting Service	X	X	X	X	X			
Search Service	X	X	X	X	X	X	X	X
Communication Services								
Recommendation Service		X	X	X			X	X
External Services						X	X	
Analytics Service	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X
File / Blob Storage								

7.4 News Feed Requirements Matrix (SRS 2.4)

Component	2.4.1 Posts	2.4.2 Updates	2.4.3 Media	2.4.4 Comments	2.4.5 Reactions	2.4.6 Feed
Web Client	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X
Profile Management Service						
Resume Service						
Job Posting Service						
Search Service						
Communication Services	X	X	X	X	X	X
Recommendation Service						
External Services						
Analytics Service	X	X	X	X	X	X
Database	X	X	X	X	X	X
File / Blob Storage			X			

7.5 Job Posting Requirements Matrix (SRS 2.5)

Component	2.5.1 Create	2.5.2 Edit	2.5.3 Delete	2.5.4 Skills	2.5.5 Details	2.5.6 Apply	2.5.7 Status	2.5.8 Filters	2.5.9 Data
Web Client	X	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X	X
Profile Management Service									
Resume Service						X	X	X	
Job Posting Service	X	X	X	X	X	X	X	X	X
Search Service	X	X	X	X			X	X	X
Communication Services									
Recommendation Service	X	X	X	X	X	X	X	X	X
External Services									
Analytics Service	X	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X	X
File / Blob Storage			X						

7.6 Student Profile Requirements Matrix (SRS 3.1)

Component	3.1.1 Student	3.1.2 Accolades	3.1.3.1 GPA	3.1.3.2 Skills	3.1.3.3 Experience	3.1.3.4 Resume
Web Client	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X
Profile Management Service	X	X	X	X	X	X
Resume Service			X	X	X	X
Job Posting Service						
Search Service						
Communication Services						
Recommendation Service		X	X	X	X	X
External Services						
Analytics Service	X	X	X	X	X	X
Database	X	X	X	X	X	X
File / Blob Storage			X	X		X

7.7 Shared Notification Requirements Matrix (SRS 3.2)

Component	3.2.1 Alerts	3.2.2 Messages	3.2.3 Push	3.2.4 Delivery	3.2.5 Tracking	3.2.6 Preferences	3.2.7 History
Web Client	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X
Authentication Service							
Profile Management Service							
Resume Service							
Job Posting Service							
Search Service							
Communication Services	X	X	X	X	X	X	X
Recommendation Service							
External Services			X	X		X	
Analytics Service	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X
File / Blob Storage							

7.8 Student Search Requirements Matrix (SRS 3.3)

Component	3.3.1 Jobs	3.3.2 Companies	3.3.3 Matches	3.3.4 Filters
Web Client	X	X	X	X
Mobile Client	X	X	X	X
API Gateway / Orchestrator	X	X	X	X
Authentication Service				
Profile Management Service		X		X
Resume Service				
Job Posting Service	X		X	
Search Service	X	X	X	X
Communication Services				
Recommendation Service	X	X	X	X
External Services				
Analytics Service	X	X	X	X
Database	X	X	X	X
File / Blob Storage				

7.9 Shared News Feed Requirements Matrix (SRS 3.4)

Component	3.4.1 Feed	3.4.2 Updates
Web Client	X	X
Mobile Client	X	X
API Gateway / Orchestrator	X	X
Authentication Service		
Profile Management Service		
Resume Service		
Job Posting Service		
Search Service		
Communication Services	X	X
Recommendation Service		
External Services		
Analytics Service	X	X
Database	X	X
File / Blob Storage		

7.10 Resume Requirements Matrix (SRS 3.5)

Component	3.5.1 Builder	3.5.2 Upload	3.5.3 Storage	3.5.4.1 Fields	3.5.4.2 Help	3.5.5.1 Submit	3.5.5.2 Review	3.5.5.3 Required
Web Client	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X
Profile Management Service			X				X	
Resume Service	X	X	X	X	X	X	X	X
Job Posting Service								
Search Service								
Communication Services								
Recommendation Service		X	X				X	
External Services								
Analytics Service	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X
File / Blob Storage	X	X	X			X	X	

7.11 Hardware Requirements Matrix (SRS 4.0)

Component	4.1 Performance	4.2 Browsers	4.3 Devices	4.4 Supported
Web Client	X	X	X	X
Mobile Client	X	X	X	X
API Gateway / Orchestrator	X			
Authentication Service	X			
Profile Management Service	X			
Resume Service	X			
Job Posting Service	X			
Search Service	X			
Communication Services	X			
Recommendation Service	X			
External Services			X	X
Analytics Service	X			
Database	X			
File / Blob Storage	X			

8.0 APPENDICES

This section is optional.

Appendices may be included, either directly or by reference, to provide supporting details that could aid in the understanding of the Software Design Document.